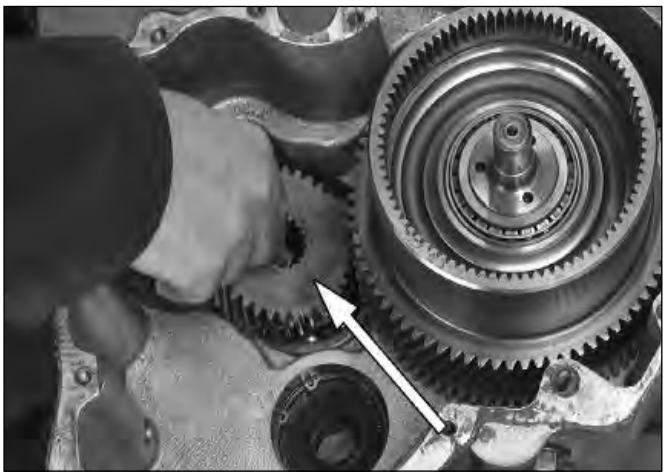
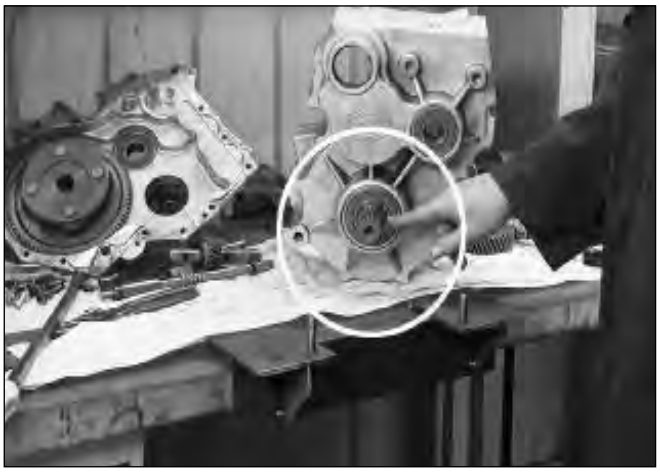
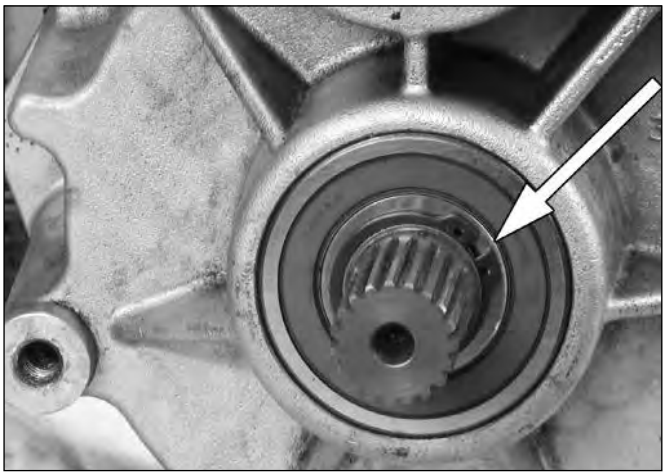
25. Place the input shaft assembly on a clean work bench.
26. Remove the 35T helical gear.

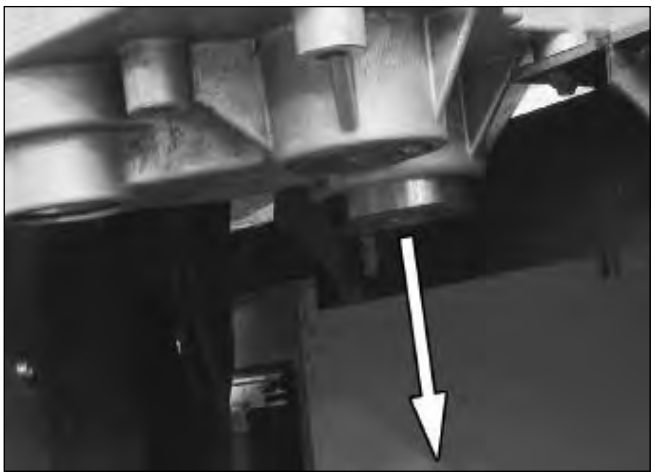
27. Remove the transmission from the stand and set it upright. Locate the retaining ring on the output shaft. Remove this retaining ring.

28. Taking the housing to a press, remove the left hand side output shaft assembly.




29. Using a drift punch, gently tap the output shaft outer bearing from the housing.

30. Remove the output shafts inner retaining ring and remove the inner retaining ring of the brake shaft assembly.

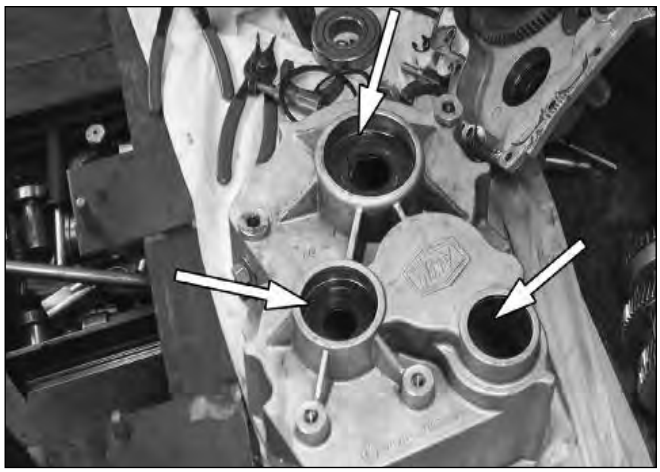


31. Remove the flanged bushing from the bearing of the idler shaft assembly.
32. Using a drift punch, gently tap the outer bearing of the brake shaft assembly from the housing.

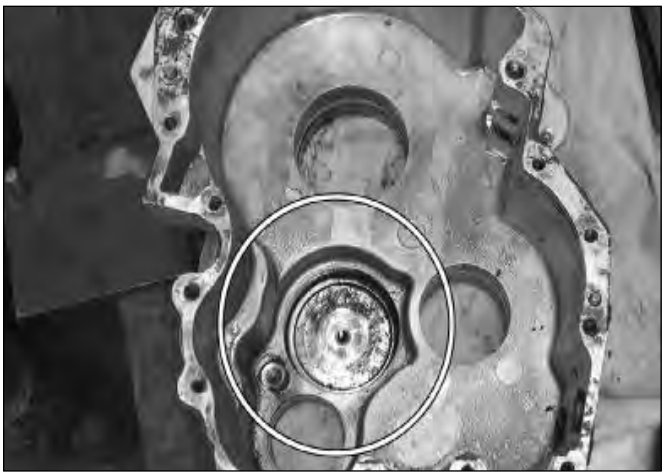
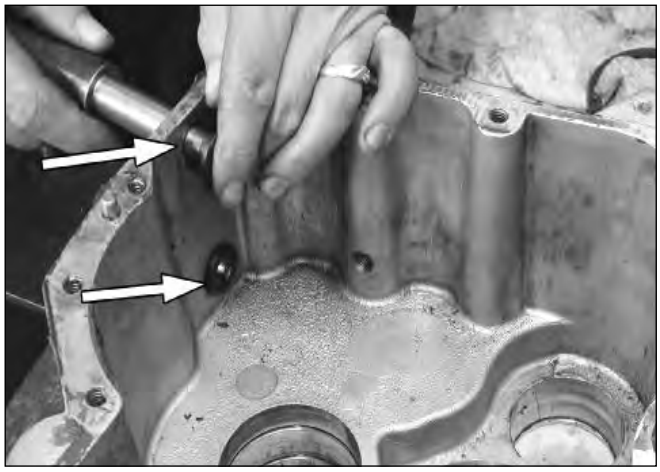
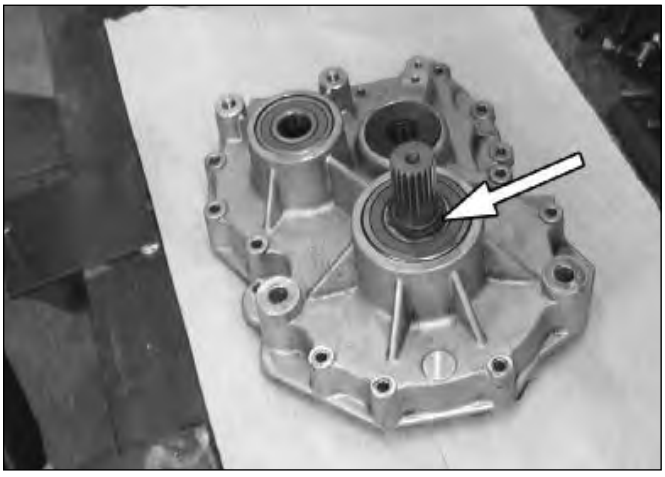



33. Flip the left hand housing over to remove the remaining outer retaining rings.

34. Remove all seals from the housing.



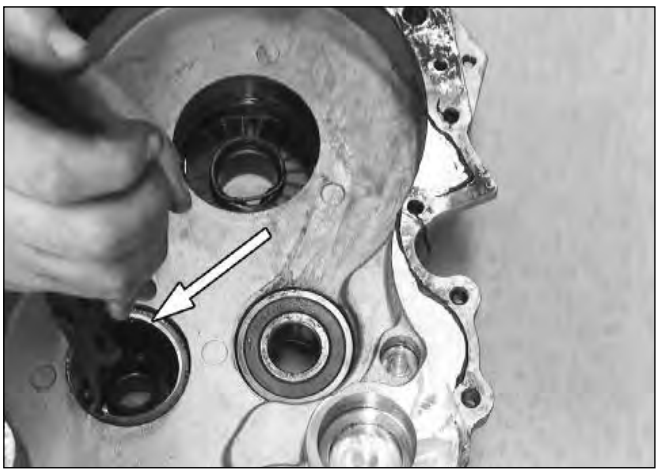

35. Remove the remaining idler shaft bearing using an internal bearing puller.
36. Remove both shift shaft bushings from the housing.
37. Locate the right hand side transmission housing and remove the retaining ring at the output shaft.

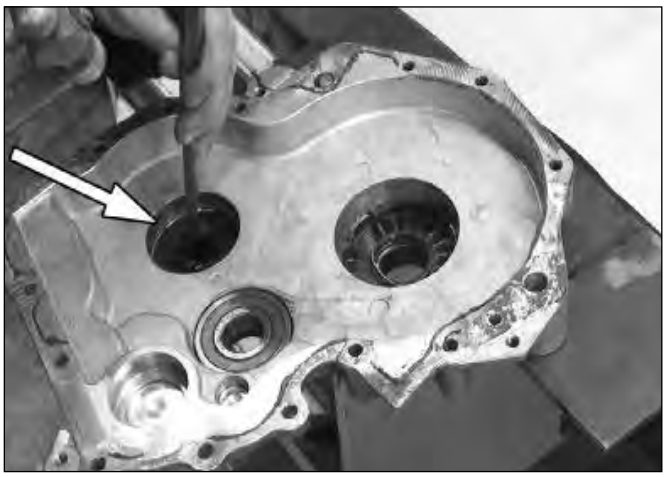





38. Take the housing to a press and push out the right hand side output shaft assembly.

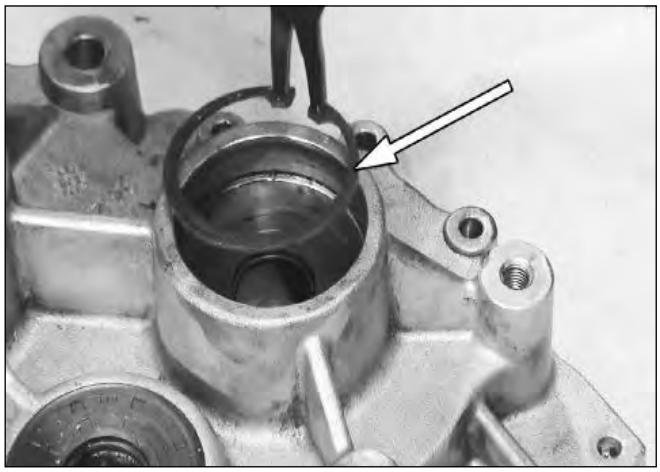


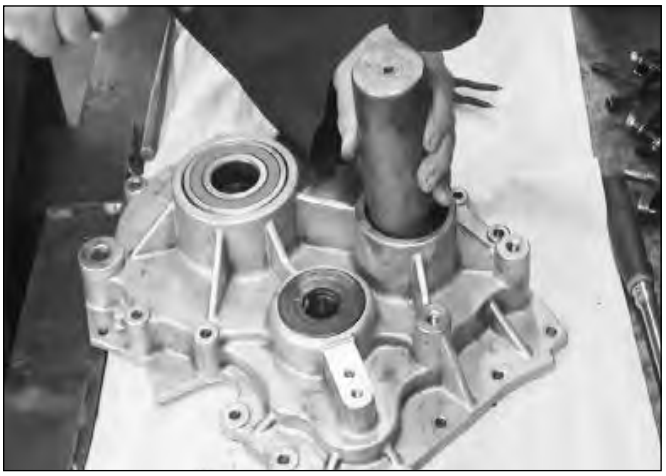
39. Remove the inner retaining ring of the brake shaft assembly.
40. Using a drift punch gently tap out the outer bearing of the brake shaft assembly.



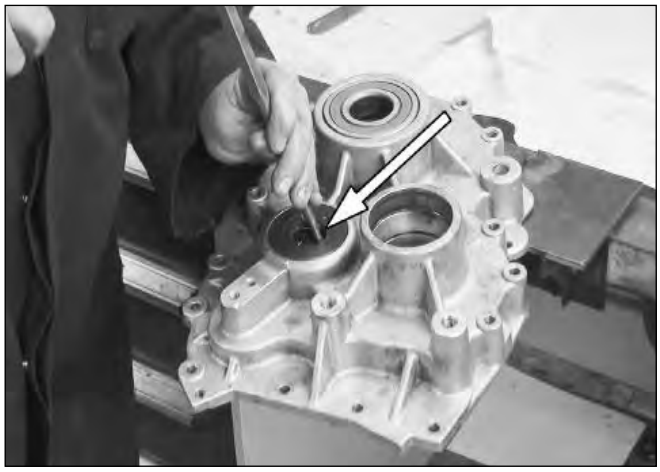
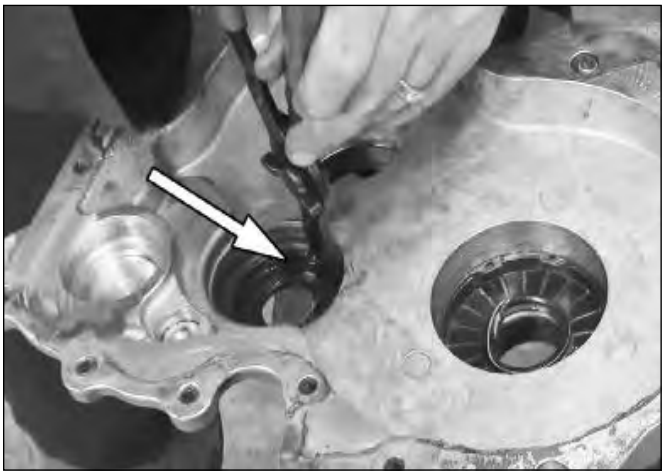
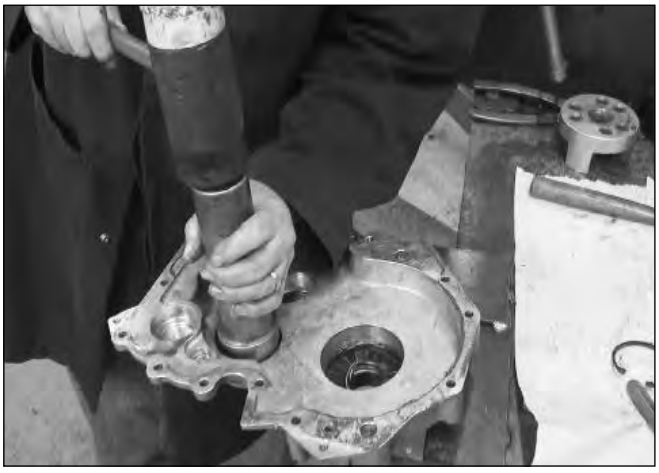


41. Turning the housing over, remove the second retaining ring at the brake shaft assembly and remove the seal.





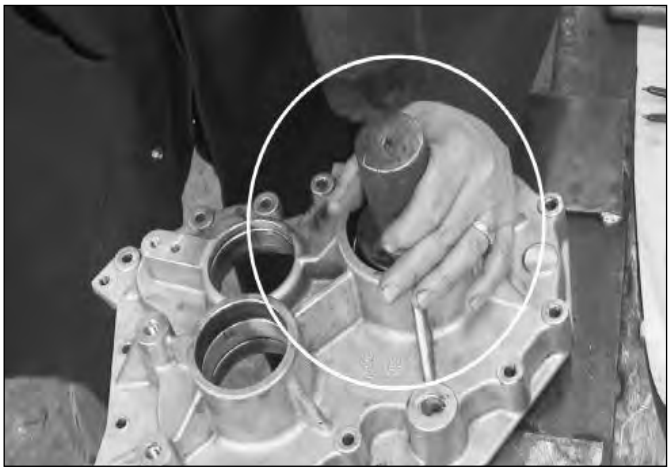
42. Place the transmission housing back on to the stand to remove the idler shaft bearing
43. Flipping the housing back over, remove the retaining ring at the idler shaft location.
44. Remove the idler shaft seal.
45. Remove the outer output shaft bearing.


46. Remove the retaining ring to the inside of the seal at the output shaft location.
47. Flip the housing over and remove the second retaining ring at the outside of the seal




48. Remove the output shaft seal.



Disassembling the Input Shaft.

1. Lay out the input shaft assembly on a clean work surface.
2. Take input shaft to press and remove bearing from the end of shaft.



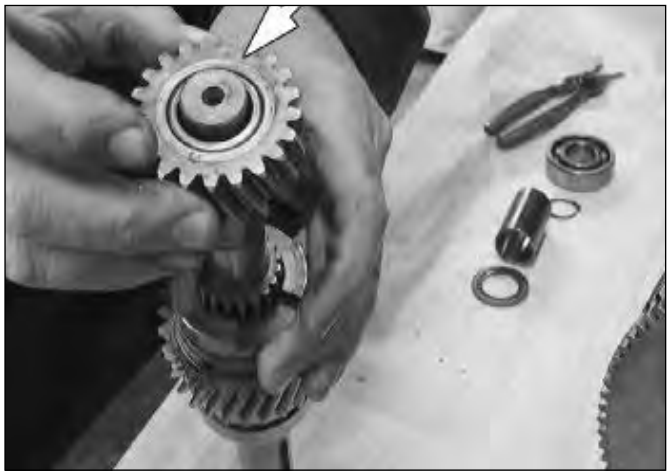


3. Remove the retaining ring located beneath the bearing and slip off the spacer.

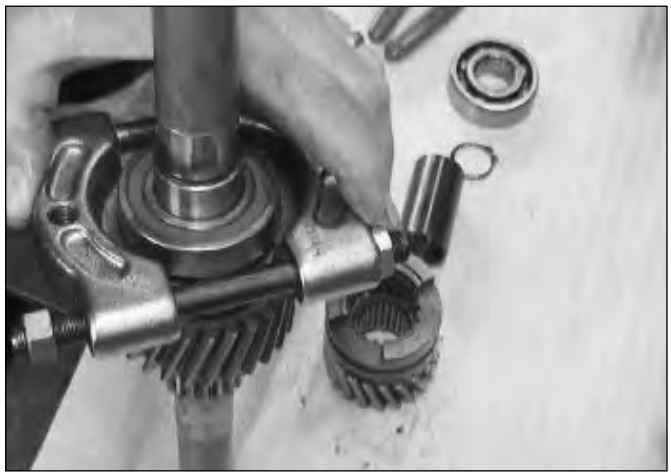


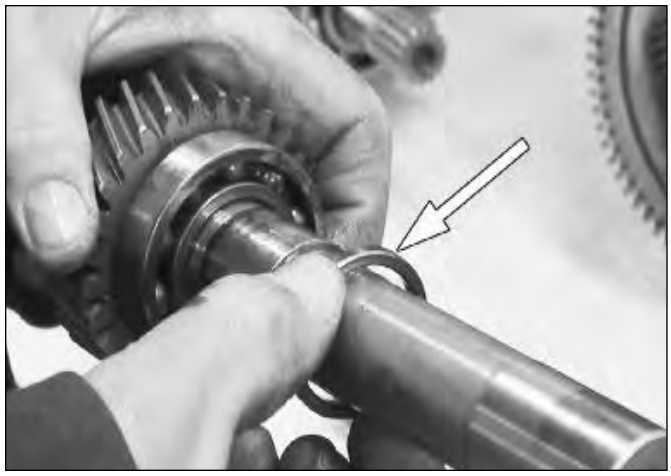


4. Remove the thrust washer.
5. Remove the reverse pinion.

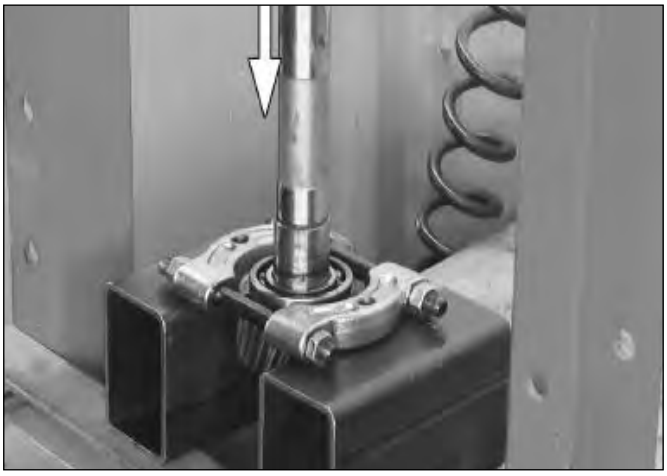
6. Slip off the forward/reverse shift collar.
7. Flip the input shaft around to remove the two bearings that are located one on top of the other. Install a bearing removal tool between the two bearings and take to the press to remove the first bearing.
8. Remove the spacer located between the two bearings.



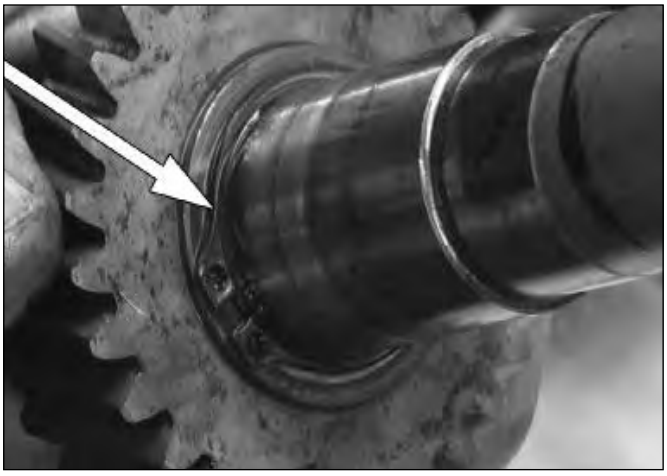
9. Place the bearing removal tool beneath the second ball bearing and use the press to remove.





10. Slide the bearing from the shaft.
11. Remove the retaining ring securing the forward pinion.





12. Slip the forward pinion off of the input shaft.

⚠ IMPORTANT

Please take note that there is a thrust washer located on both sides of the forward pinion.



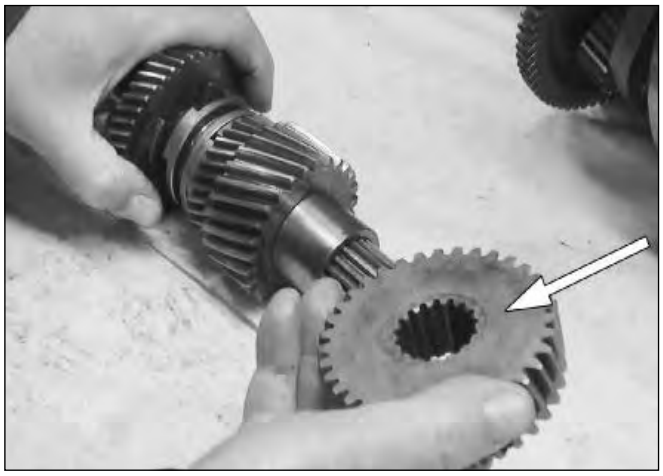
Complete disassembled input shaft.

Clean all components in a varsol bath or an environmentally safe solution before Inspecting for any damaged or worn parts.

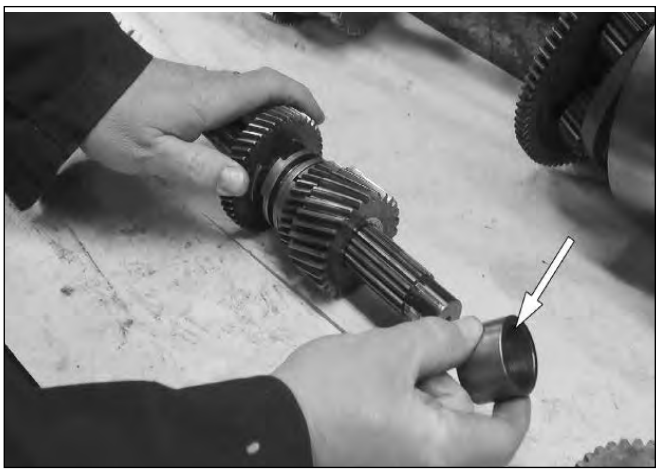
Disassembling the Idler Shaft Assembly

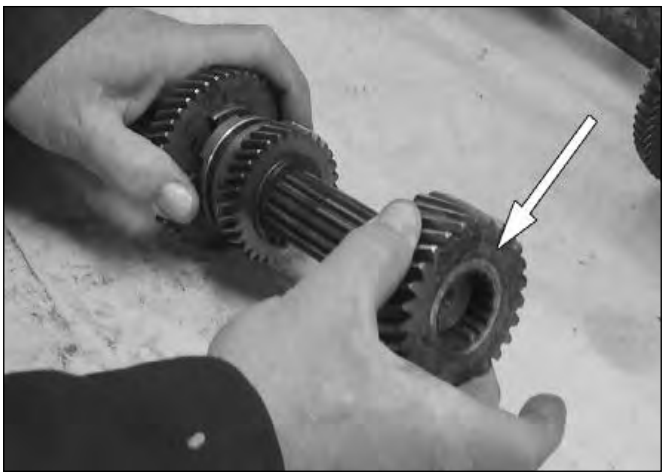
1. Lay out the idler shaft assembly on a clean work surface.
2. Remove the 35 tooth helical gear.



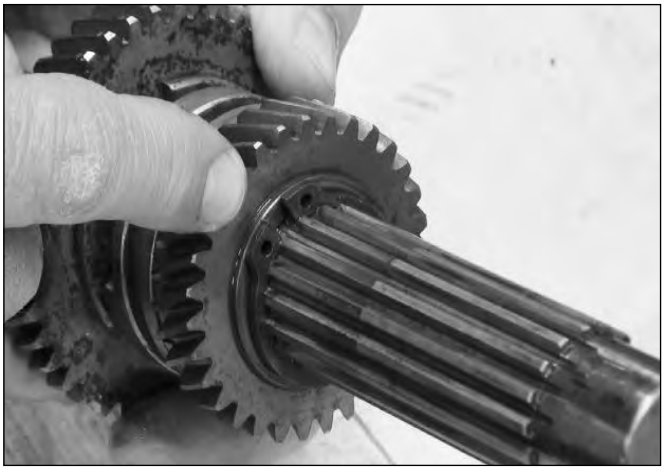


3. Remove the spacer. .
4. Slide off the 27 tooth helical pinion and note the machined recess on the one side.

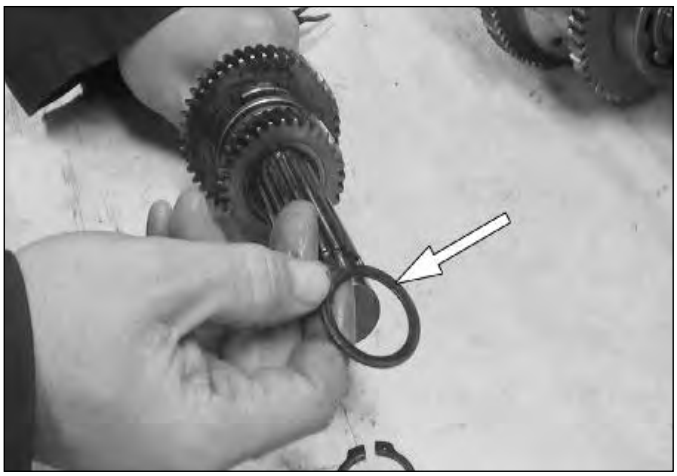




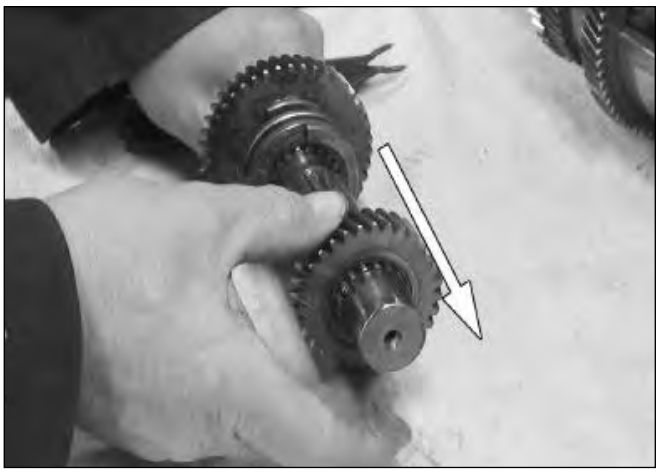
5. Next locate the retaining ring securing the 31 tooth helical pinion.
6. Remove the retaining ring.

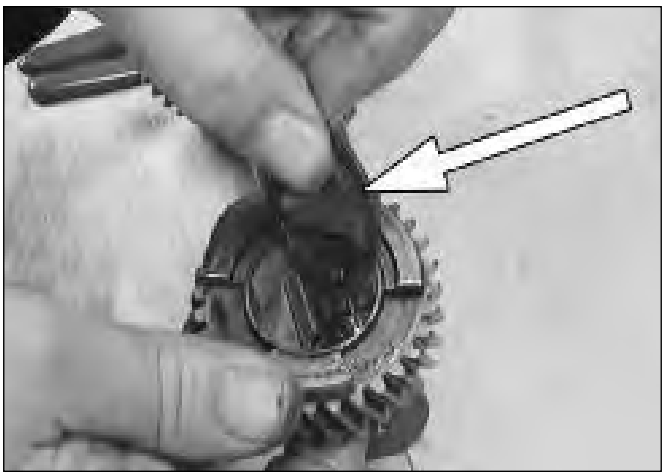


7. Slide off the thrust washer.



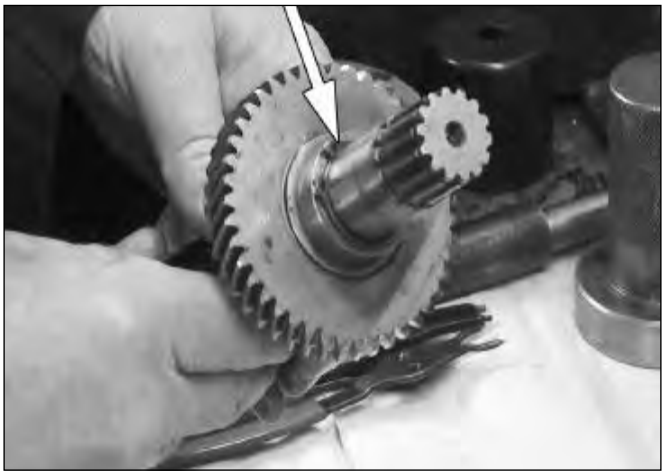
8. Slide off the 31 tooth helical pinion. Take note of the thrust washer located to the dog tooth side.



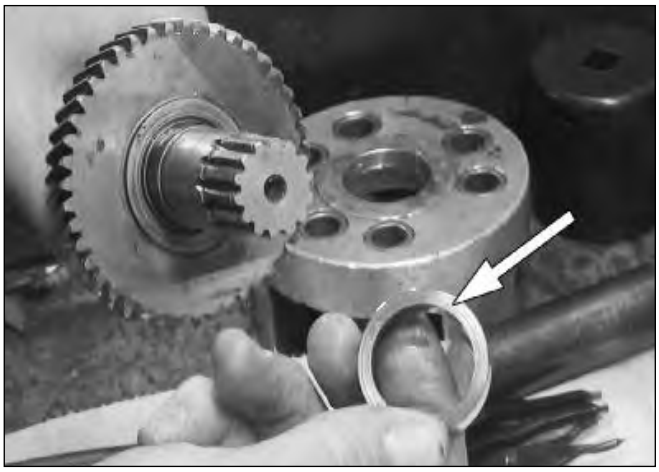


9. Slide the hi lo/shift collar from the idler shaft.
10. Flipping the idler shaft to the other side, locate and remove the retaining ring securing the spacer below.





11. Remove the spacer.
12. Slide the 45 tooth helical pinion from the idler shaft.



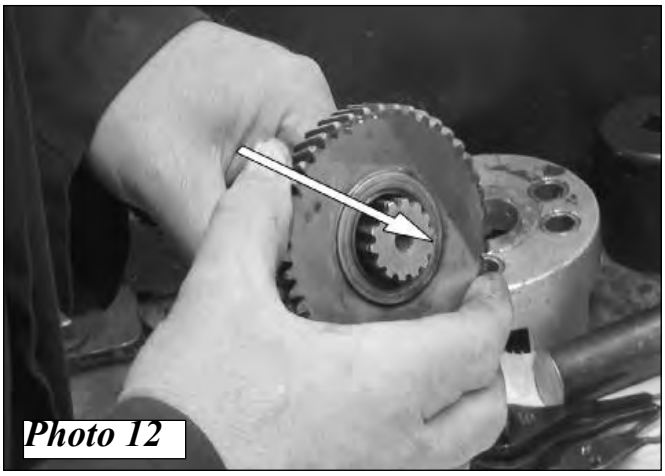
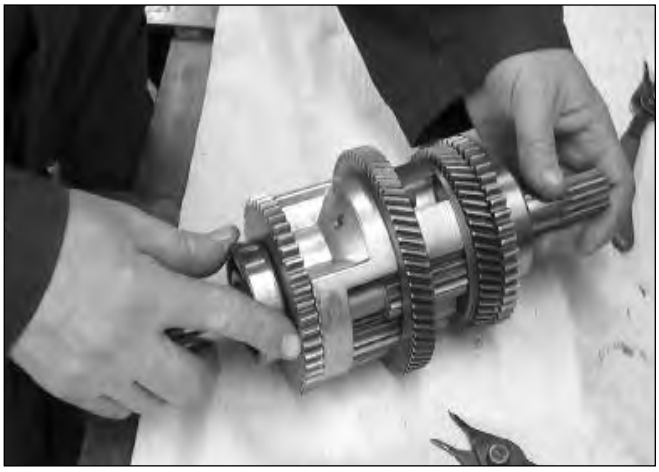


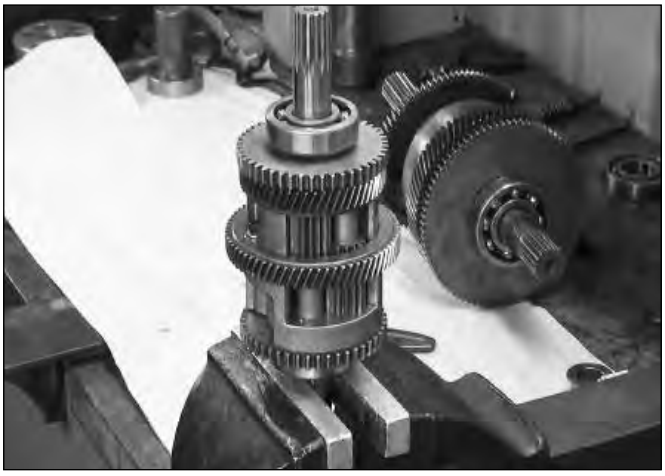
Photo 12

Clean all components in a varsol bath or environmentally safe cleaning solution and lay out on a clean work bench for inspection.

Disassembling the Brake Shaft Assembly (Steering Differential Assembly)

1. Set the brake shaft assembly to a clean work surface. .
2. Secure the differential into a aluminum jawed vice.

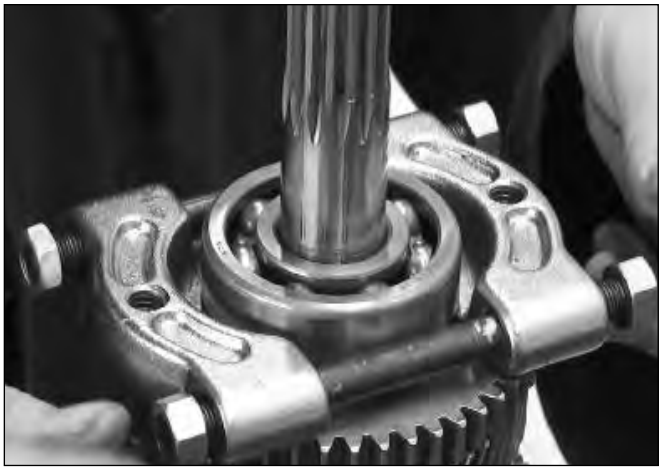




3. Install a bearing removal tool to remove the ball bearing from the one end of the differential. *See following page.*

NOTE

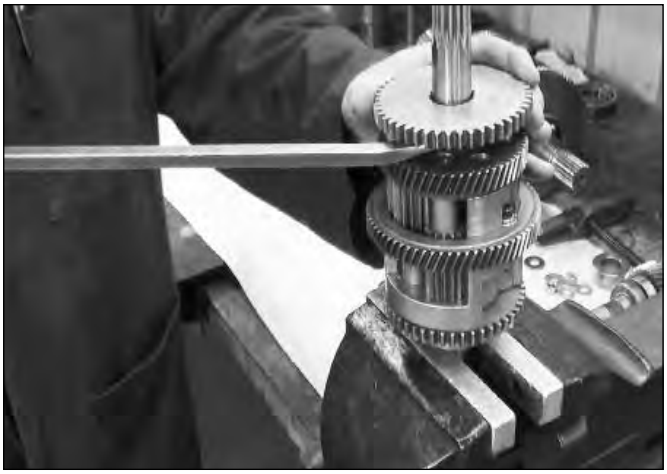
There is a the shim washer beneath the ball bearing.





4. Remove the 46 tooth spur pinion. Gently pry off with a pry bar.
5. Remove the thrust washer located below the spur pinion.

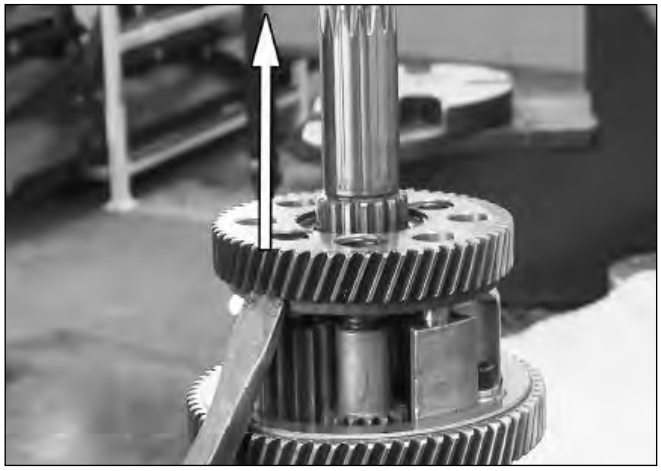








6. Remove four fasteners securing the 55 tooth helical gear.
7. Using a pry bar gently pry the helical gear from the differential cage. Note the dowel bushings at each fastener location.

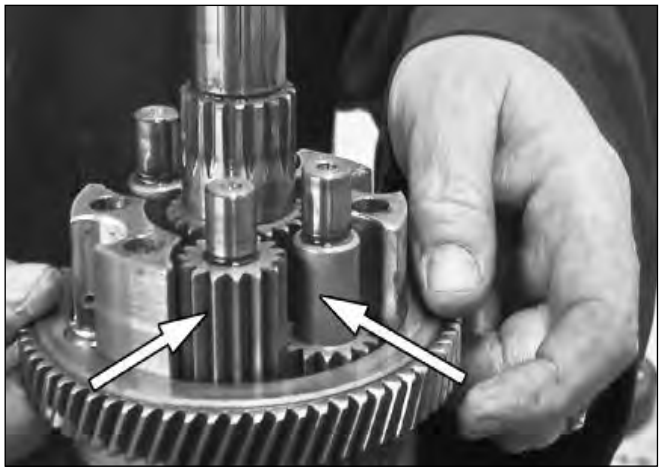


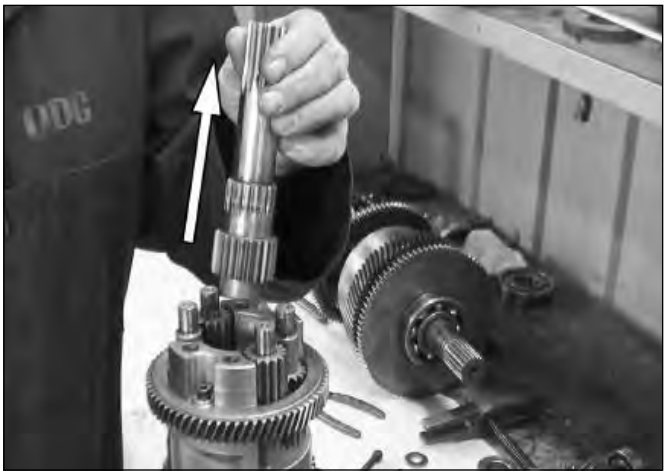


NOTE

Each planet gear is staggered one up and one down.

8. Pull up and remove the brake shaft.





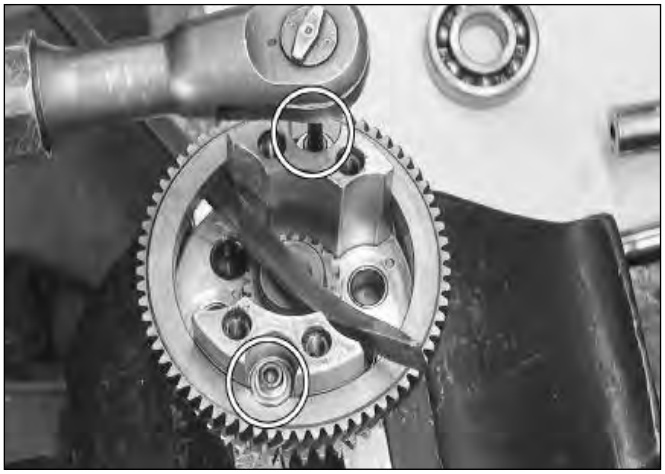
9. Remove each planetary gear.



10. Remove the two fasteners securing the 67 tooth helical gear to the differential cage.

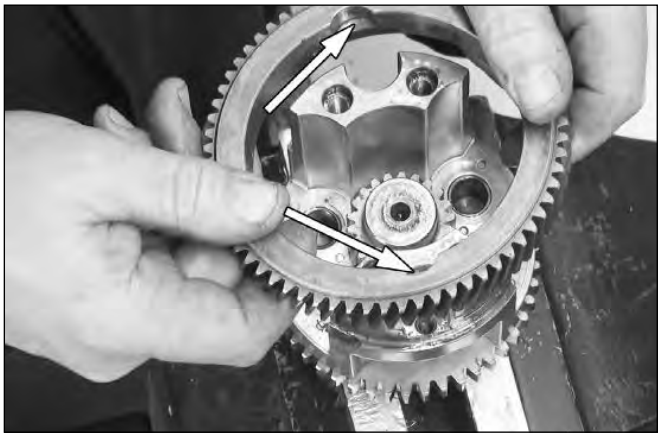
NOTE

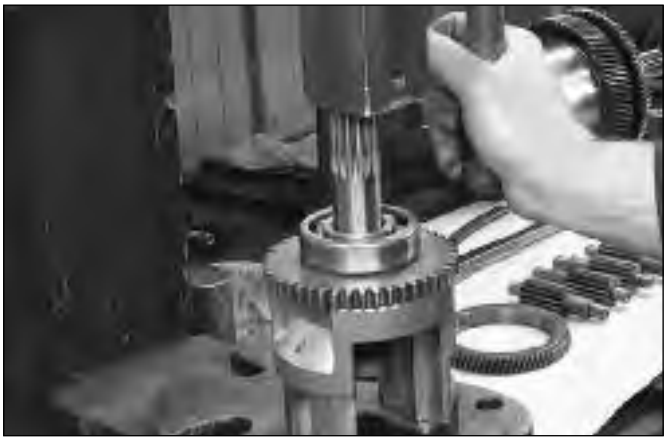
Each fastener has a dowel bushing and lockwasher.





11. Remove the 67 tooth helical gear. Take note of the two machined areas on the gear 180 degrees from each other. This will be important when the differential is reassembled.
12. Press the second brake shaft (located on the differential cage side), out of both ball bearing and 46 tooth spur pinion below.







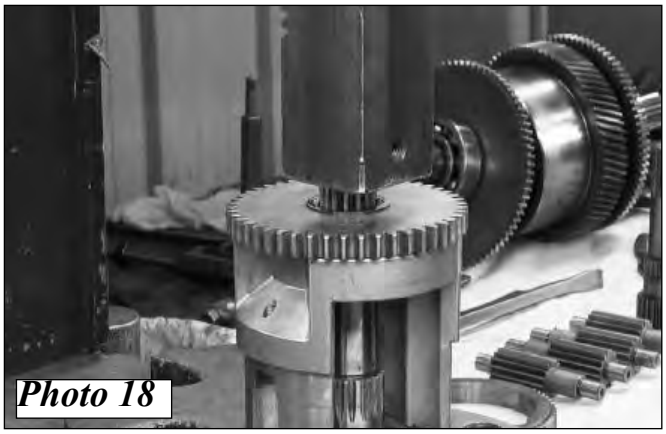
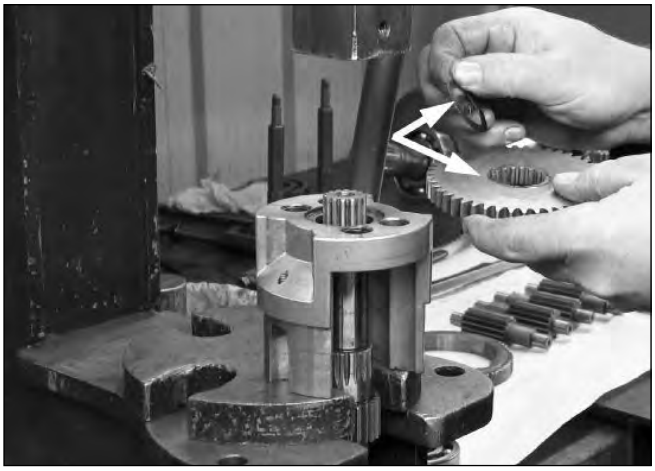


Photo 18

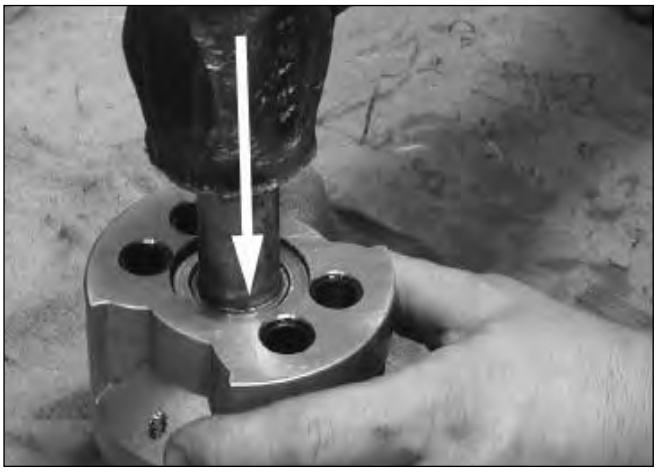
IMPORTANT

Take note of the shim washer above the spur pinion.

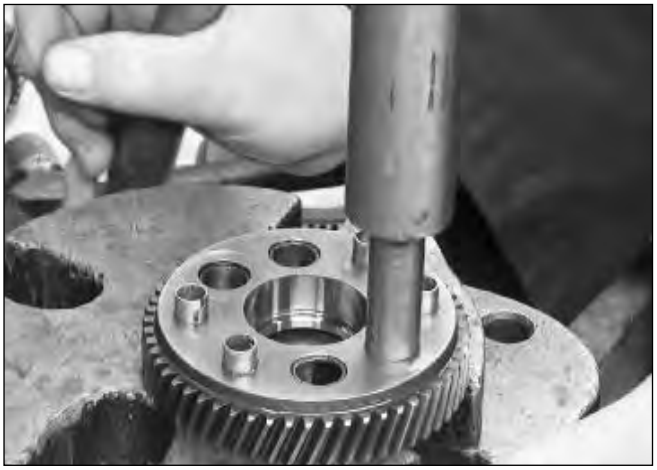
13. Remove the thrust washer from the cage below.




14. Remove the needle bearings from both differential cage and 55 tooth helical gear plate using specialized bearing remover tool P/N 608-54.




15. Press out each of the four oilite bushing from the 5 tooth helical gear.
16. Press out each of the four oilite bushing from the differential cage.




17. Wash all components is a varsol bath or environmentally friendly solution and lay out for inspection.

Disassembling the Reverse Idler Shaft

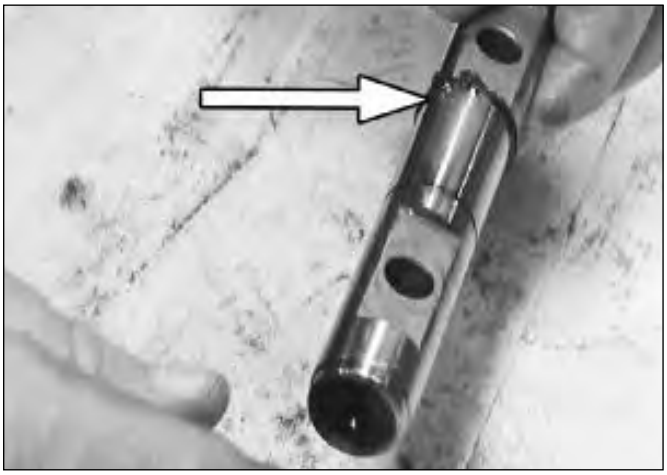
1. Remove the retaining ring securing the 20 tooth idler pinion on the reverse shift shaft.
2. Slide the idler pinion off of shaft along with the thrust washer installed above it.



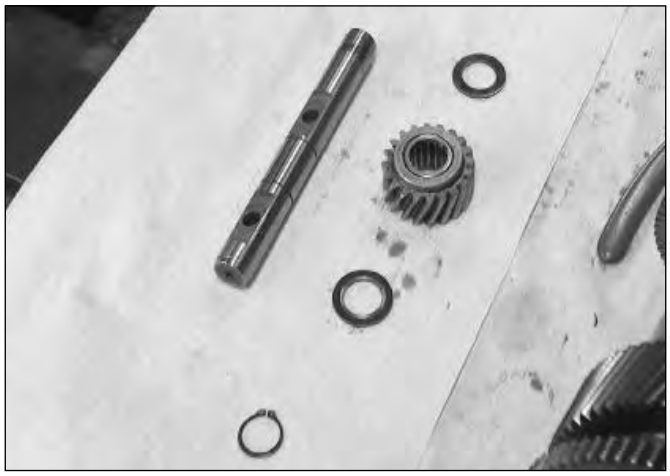


3. Remove the thrust washer from the shaft located behind the idler pinion.
4. Remove the retaining ring.



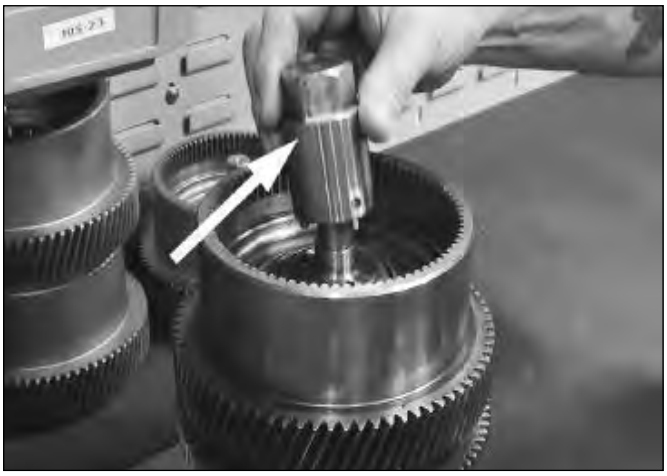


5. Clean all components in a varsol bath or environmentally friendly cleaning solution and lay out on a clean work bench for inspection.



Disassembling The Left Hand Side Output Shaft Assembly

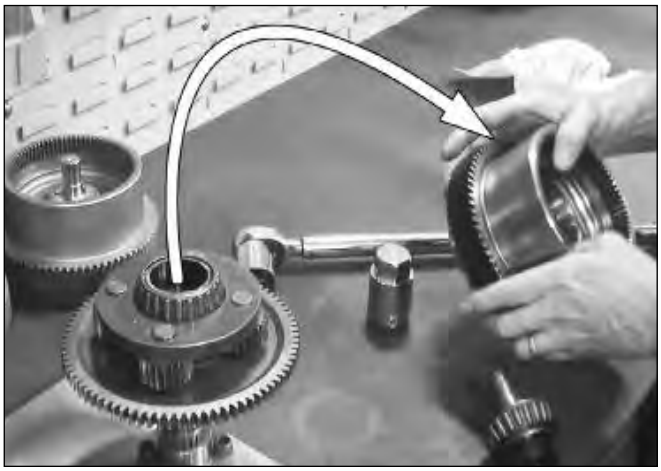
1. Place the left hand side output shaft assembly into the assembly/disassembly fixture and remove the planetary stub shaft. This stub shaft requires a specialized socket tool to unthread it from the output shaft.

2. Pull the stub shaft and tapered needle from the assembly.

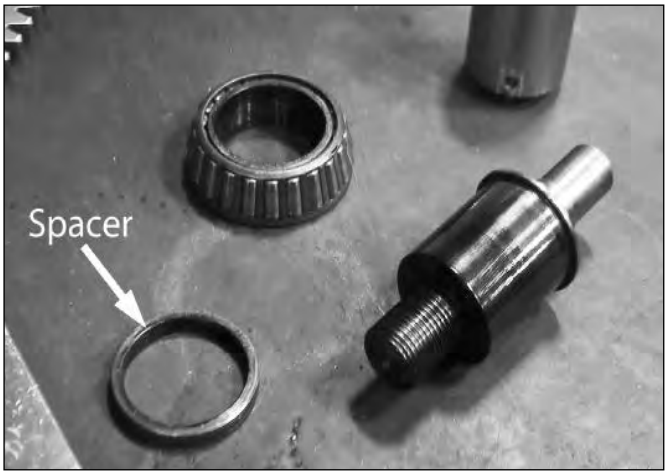



3. Remove the planetary hub.
4. Remove the second tapered bearing.



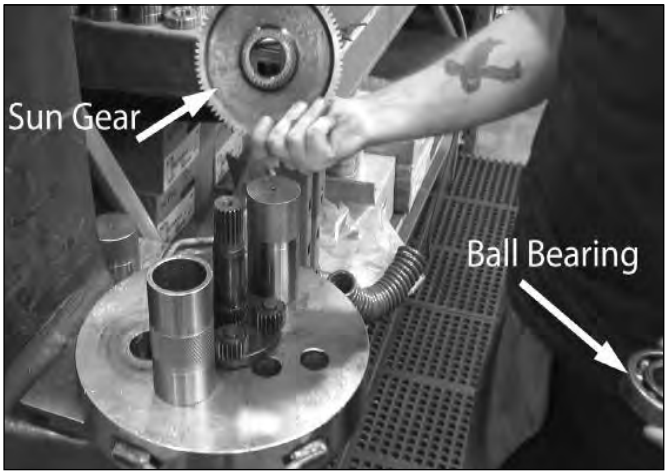

5. Slip the tapered bearing off the previously removed stub shaft and separate the spacer from the bearing.



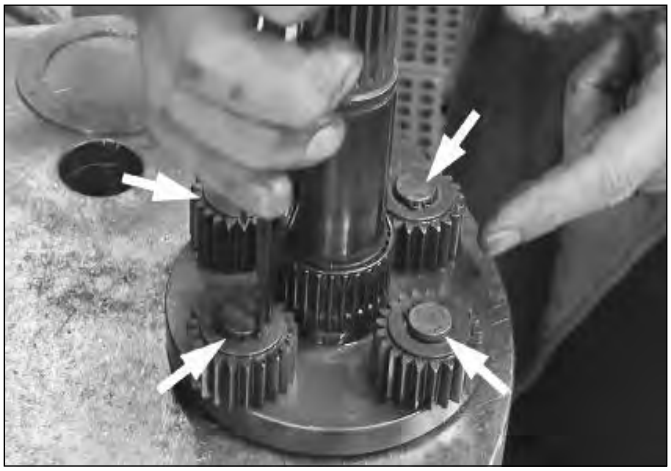


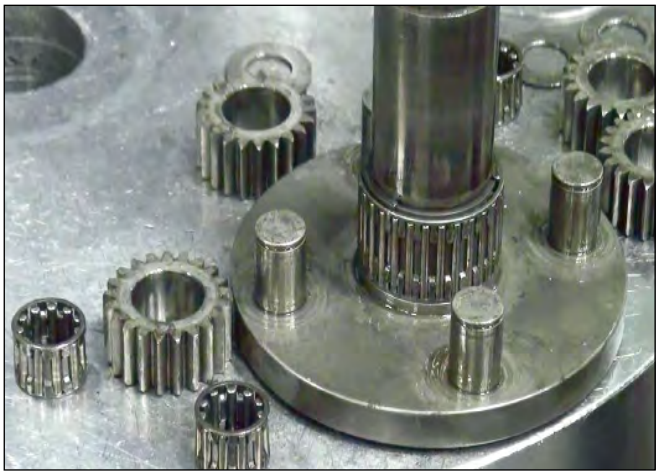
6. Press the output shaft from the sun gear and ball bearing.



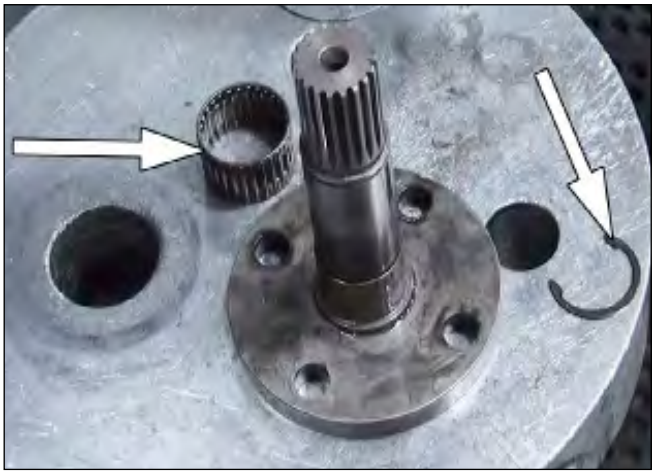


7. Locate each of the 21 tooth planet gears. Remove the retaining rings that secure each of them to the planet carrier pins.






8. Slip each planet gear off the carrier pins and set aside each flat washer located to the top of each one. Remove each needle bearing from the planet carrier pins.
9. Place the output shaft to the arbor press and remove each of the planet carrier pins.
10. Remove the retaining clip above the needle bearing and slip the needle bearing off of the output shaft.

Disassembling The Right Hand Side Output Shaft

1. Locate the right hand side output shaft assembly and repeat steps 6 through 11 of **Disassembling The Left Hand Side Output Shaft Assembly**

Admiral Transmission Assembly

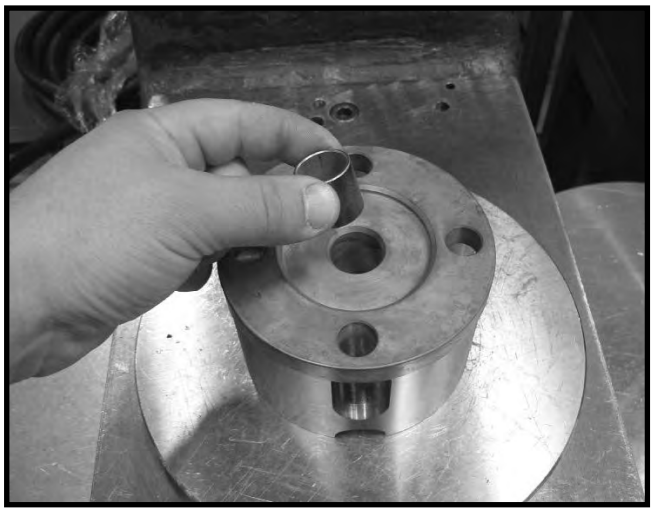
RIGHT OUTPUT SHAFT ASSEMBLY



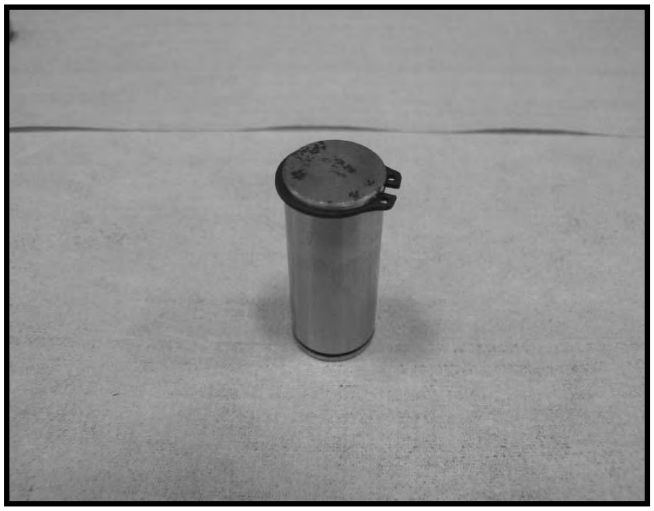



1

Position Output Shaft on special fixture on press.

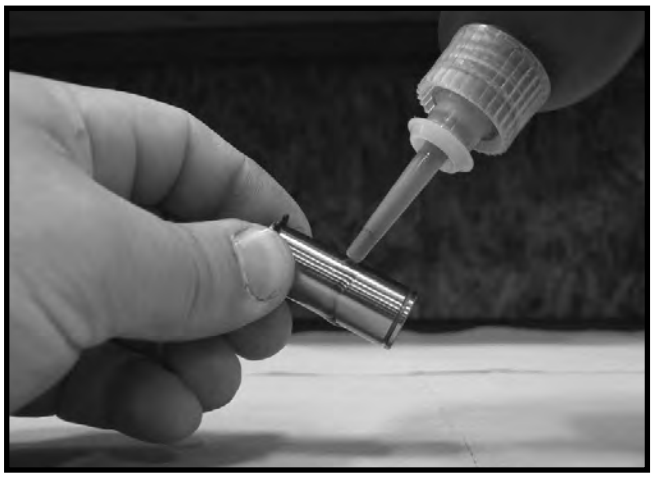



Install **105-26 Bushing** into end of shaft using a special fixture. ***Note:*** **Bushing must go in all the way to the bottom of the bore.**

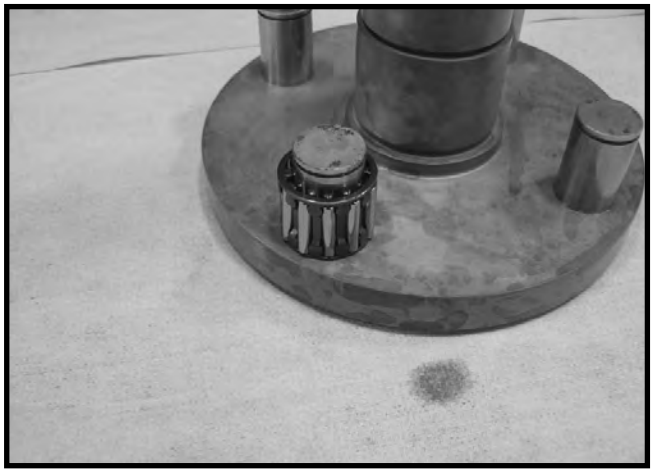
Install (1) **106-22 Retaining Ring** on end of **34-225 Pin**.

Apply 60941 Loctite on middle of the pin and press through the output shaft until the retaining ring touches the shaft. ***Note:*** **If pin is pressed too hard against the shaft it will crack.** Repeat for other three pins. Wipe off excess loctite.

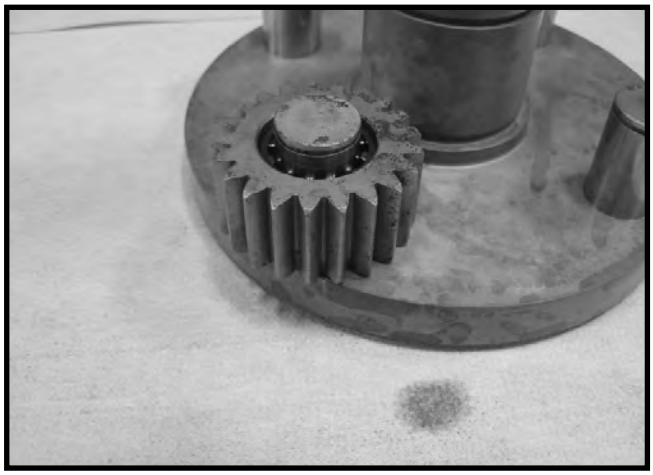


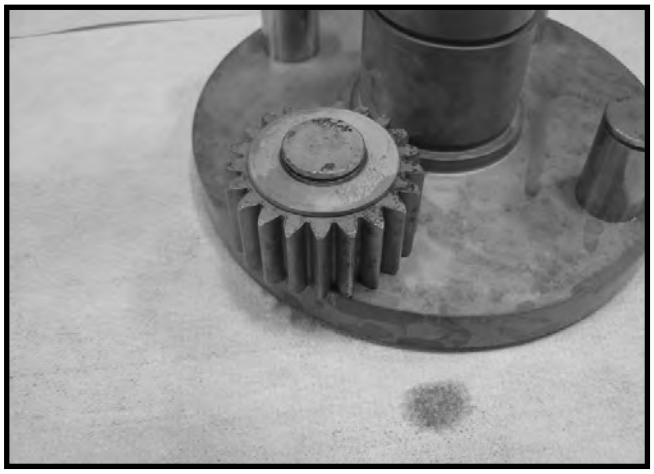


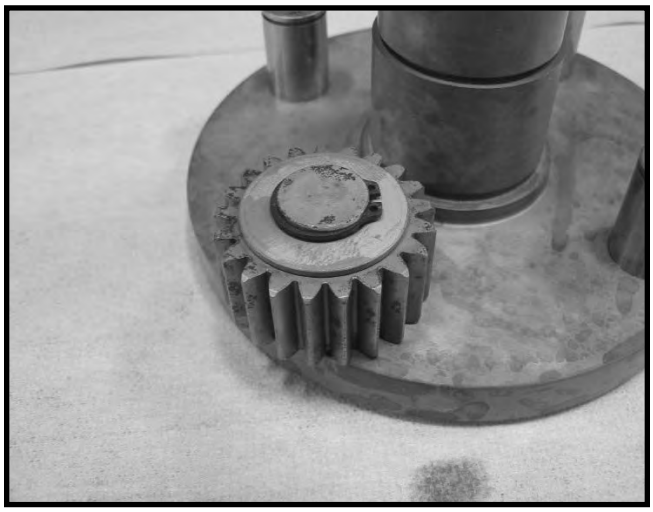


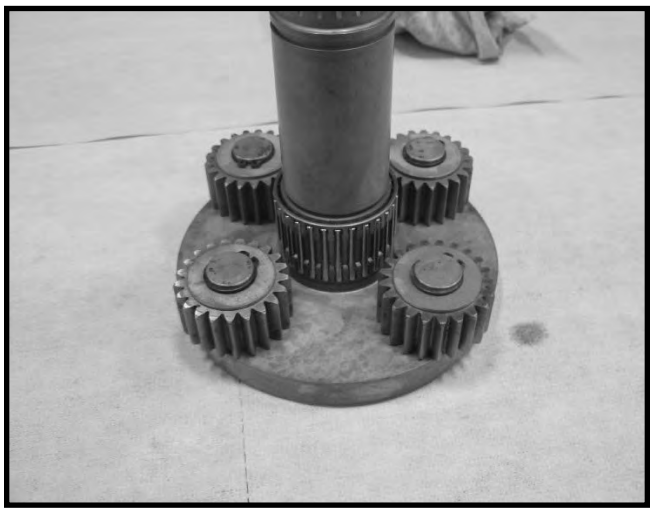


Install **101-82 Needle Bearing** on pin, followed by **34-217 Planetary Gear** with the step downwards, **108-93 Washer** and **106-22 Retaining Ring.** Repeat for other three pins.

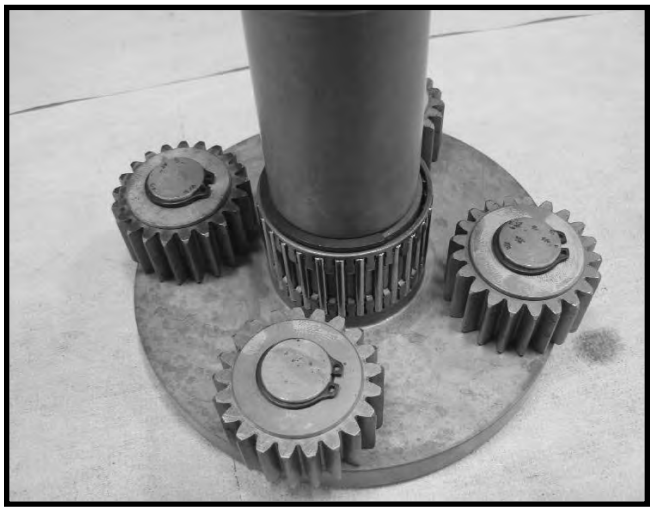


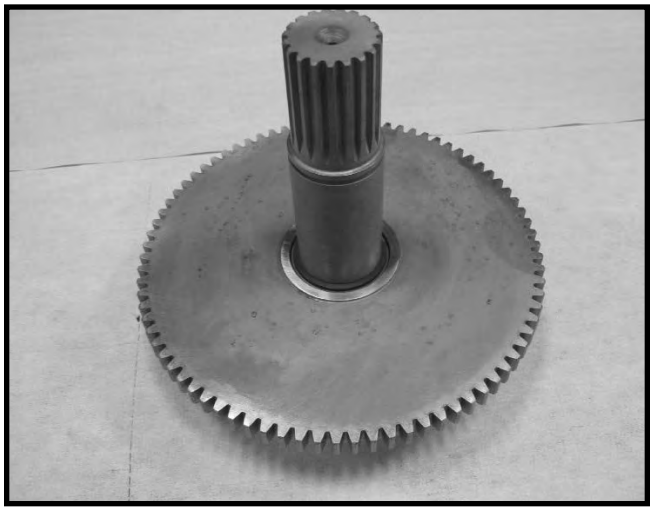


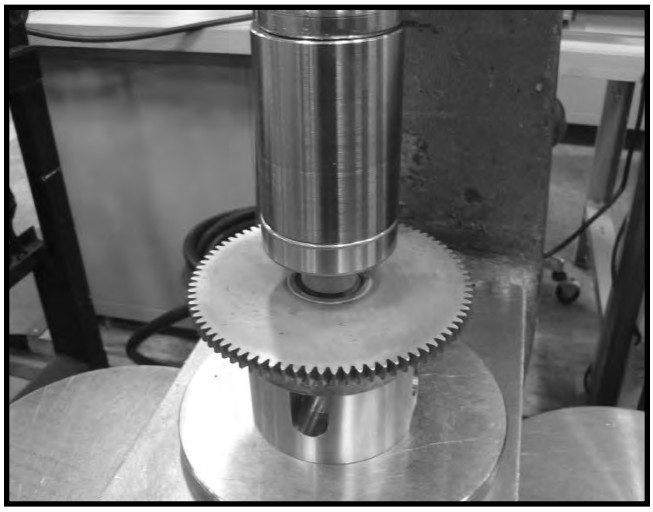


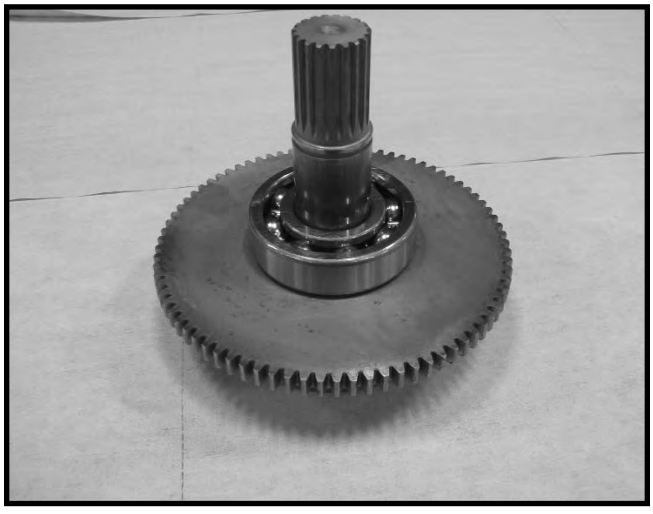


Install **101-83 Needle Bearing**, **106-31 Retaining Ring**, **34-207 Spur Gear,** and **101-73 Ball Bearing, Note: Once pressed together, 34-207 Spur gear should spin freely.**



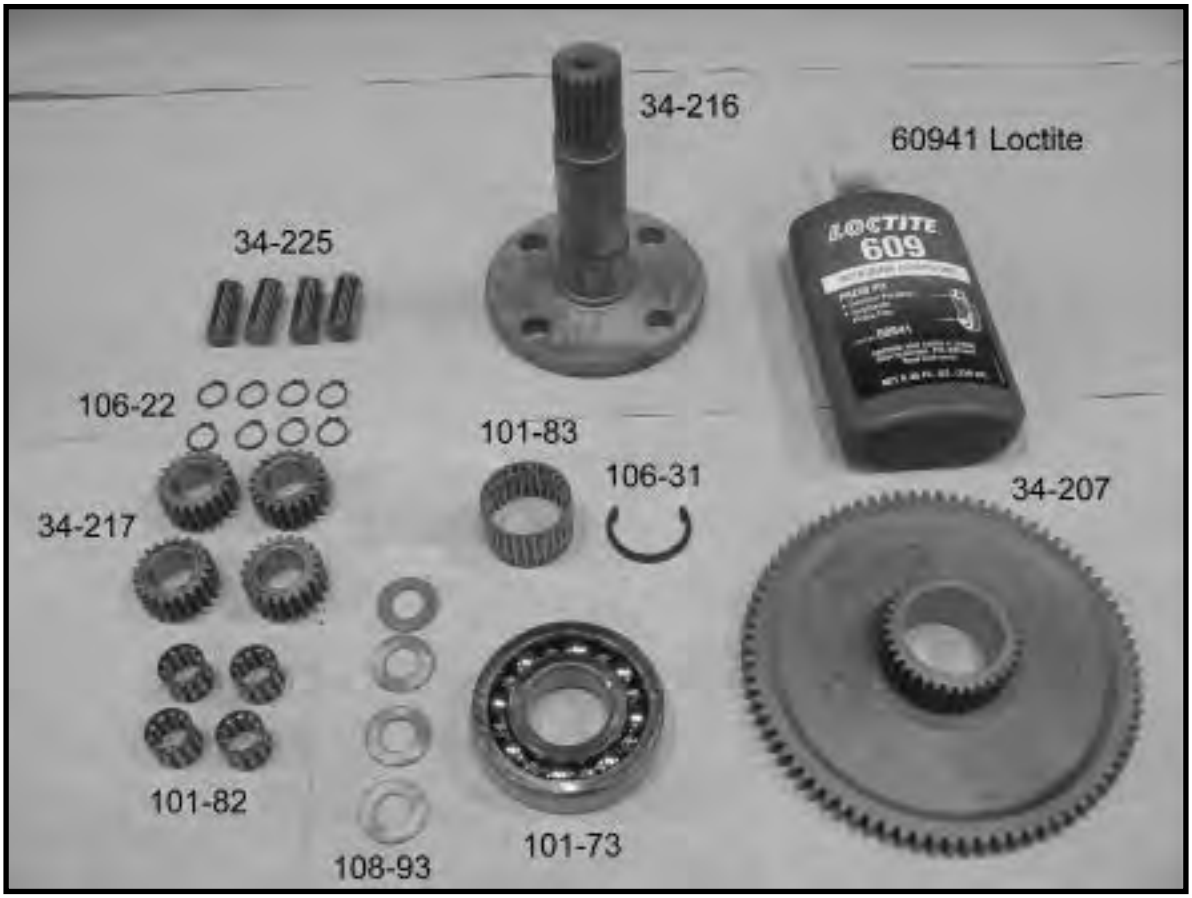






LEFT OUTPUT SHAFT ASSEMBLY

Assemble same as right side. Using **34-216 Output Shaft** instead of **34-226 Output Shaft** and no **105-26 Bushing**.





Components: 101-76 Tapered Roller Bearing, 114-22 Spacer, 34-218 Planetary stub shaft, 34-219 Planetary Hub, 34-216 Output Shaft





Press **101-76 Bearing Cups** into the **34-219 Planetary Hub** using a special fixture. ***Photo 1. Note:*** **Bearing Cups need to be installed with cup facing upwards so that the bearing cones can be installed later.**





Slide (1) **101-76 Bearing Cone** onto **34-218 Stub Shaft** with cone facing threaded side.

Take the output shaft to the vise, clamp with shaft facing down. (Make sure you use aluminum vise attachments).

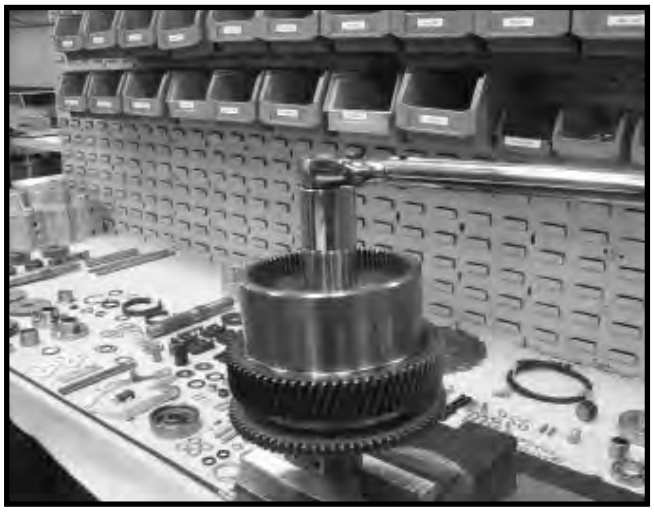




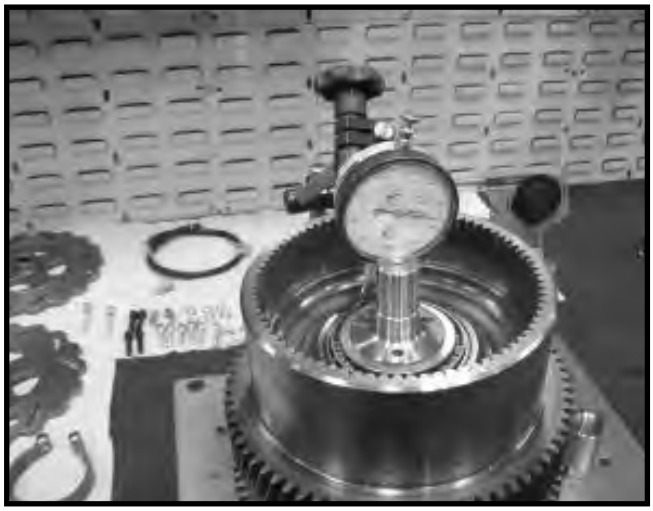
Place bearing on top of **34-216 Output Shaft** with cone facing upwards. Place **114-22 Spacer** on top.

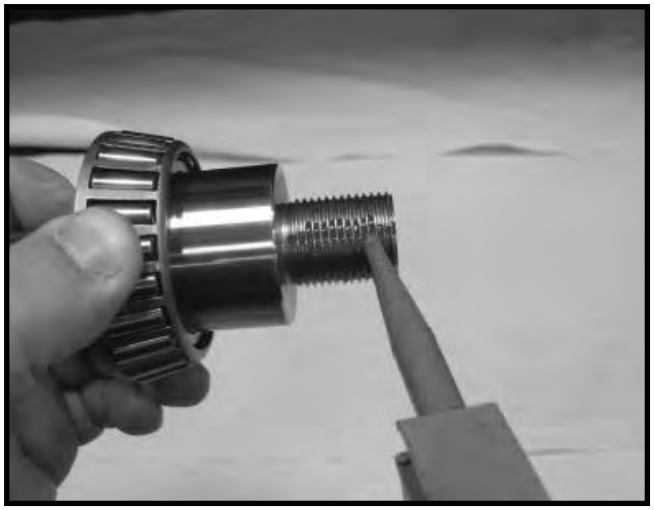
Place planetary hub with outside gear facing down.





Thread **34-218 Stub Shaft** with **101-76 Bearing Cone** into the **34-216 Output Shaft**. Torque to 100 ft lbs.





Check endplay with dial indicator (.0005 to .0025) If endplay is not enough, use required shim size.

When endplay is ok, pull apart **34-218 Stub Shaft** and apply Red 2760 LOCTITE on threads of both stub shaft and internal threads of Output Shaft.

Re-install and re-torque.



Upper Differential Assembly



Components
101-84 Needle Bearing, 101-40 Ball Bearing, 103-69/P2 Bolt 105-23 Dowel, 105-27 Bushing, 105-28 Dowel bushing, 108-03/P2 Lockwasher, 108-94 Thrust washer, 112-64 Bolts, 114-16 Shim, 34-201 Brake shaft, 34-202 Planet gear, 34-203 Spur pinion, 34-204 Helical gear, 34-205 Helical gear, 34-206 Differential Cage

Install **105-27 Bushings** into **34-204 Helical Gear** and **34-206 Differential Cage** from the inside of both the helical gear and cage. This can be identified by having a big counter bore for the **101-84 Needle Bearing** which can also be installed with the writing on the bearing facing upwards.





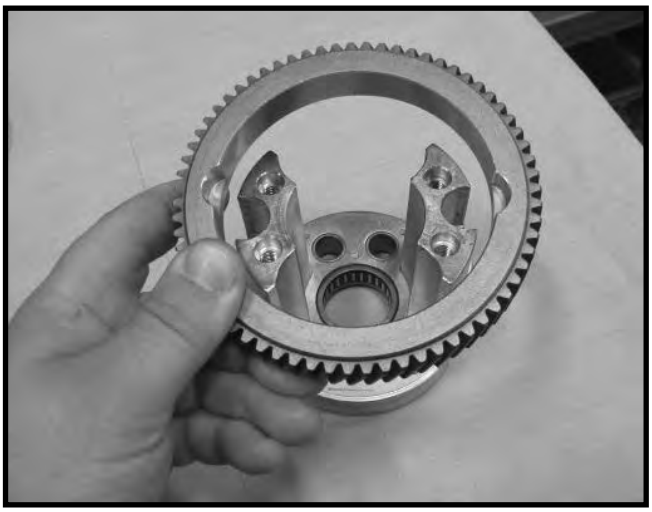




Install **101-84 Needle Bearing** into **34-206 Differential Cage**

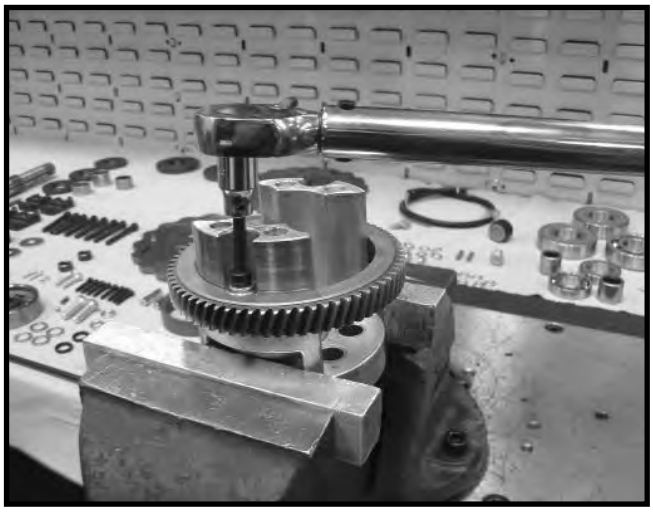
Slide **34-205 Helical Gear** onto differential cage with inside slots facing up.

Install **105-28 Dowel Bushings** into slots on helical gear and cage.

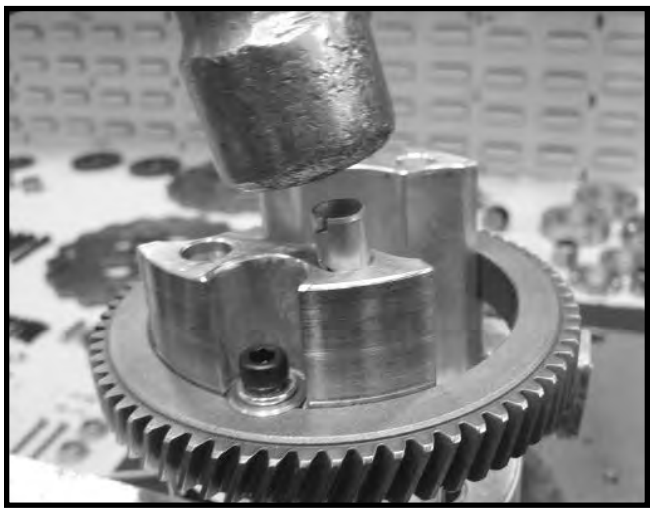


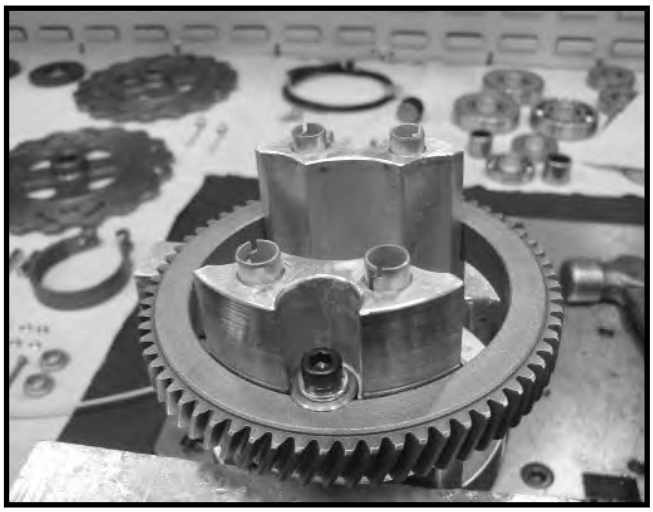




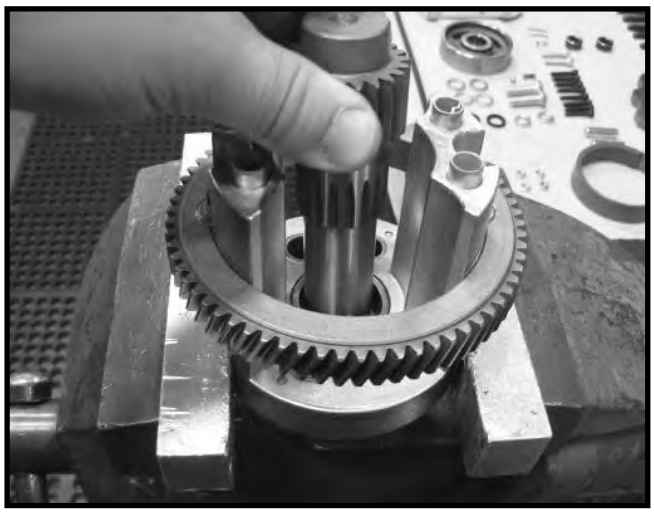


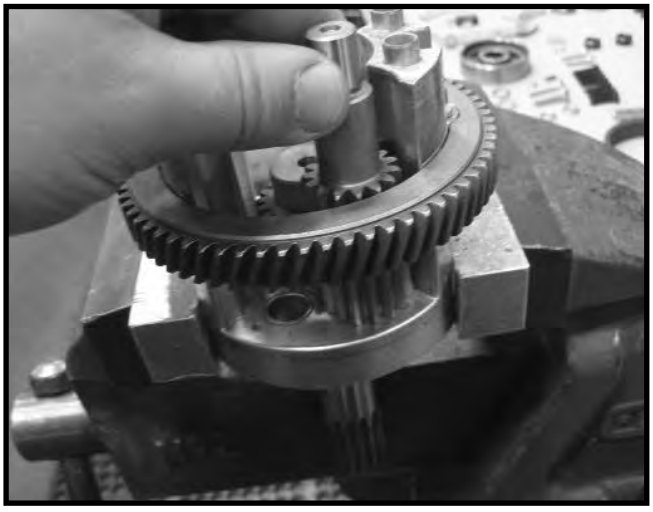
Apply 2760 Red LOCTITE to the threads of both **103-69 Bolts** and to the threads of the mounting holes in the aluminum cage where they will be installed, and install with **108-03 Lockwashers**. Torque bolts down to 16 ft-lbs



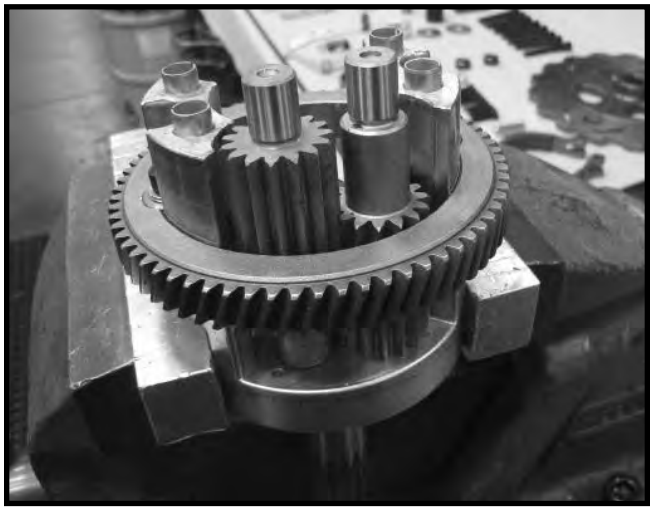


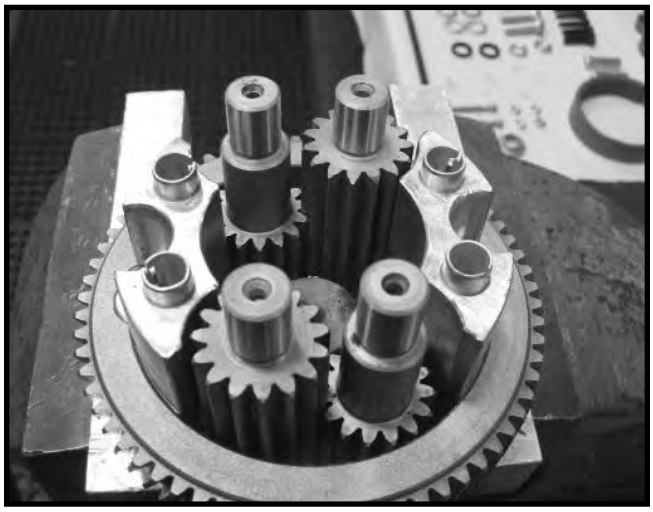
Install **105-23 Dowels** into top of cage by gently tapping with a hammer.





Install **34-201 Brake Shaft** in the **34-206 Differential Cage,** followed by the **34-202 Planetary Gears** which need to be staggered (one up and one down).

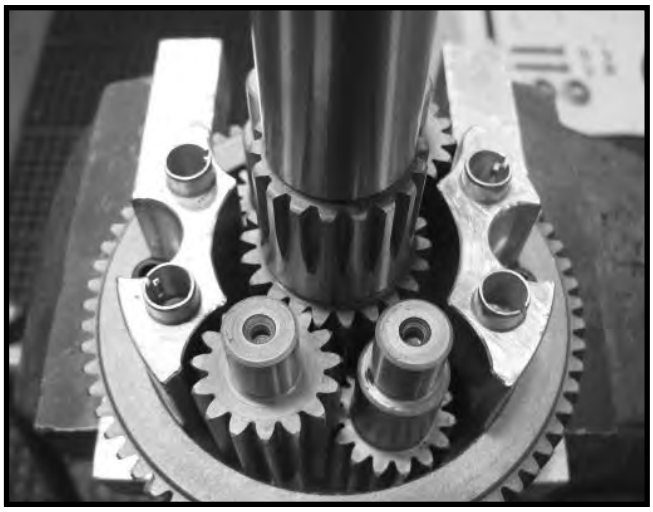




Install another **34-201 Brake Shaft**.

Note: **When installing the second 34-201 Brake shaft, shaft must be centered in the cage otherwise next step will be difficult.**





Drop on the **34-204 Helical Gear** and align the bushings with the **34-202 Planetary Gears**. The **34-204 Gear** should drop on half way. Pick up the whole assembly and spin the bottom **34-201 Brake Shaft** to insure proper assembly.

Press down evenly on **34-204 Helical Gear** until it is seated against the differential cage. **.**

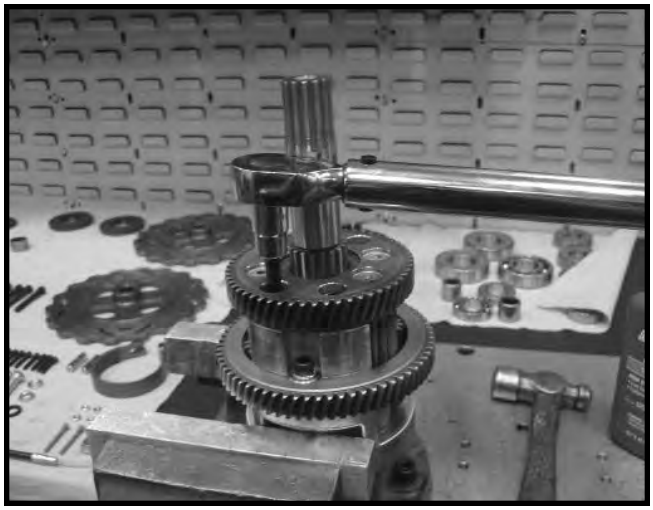






Note: **Any resistance due to parts being too tight, do not force. If this occurs, disassemble and check parts for interference.**



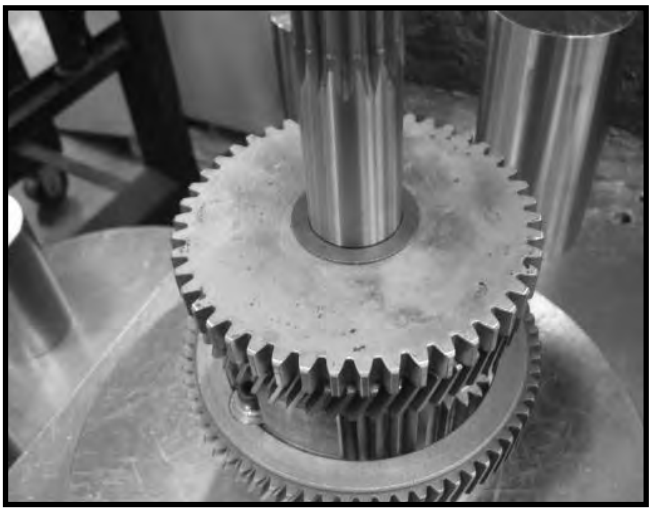


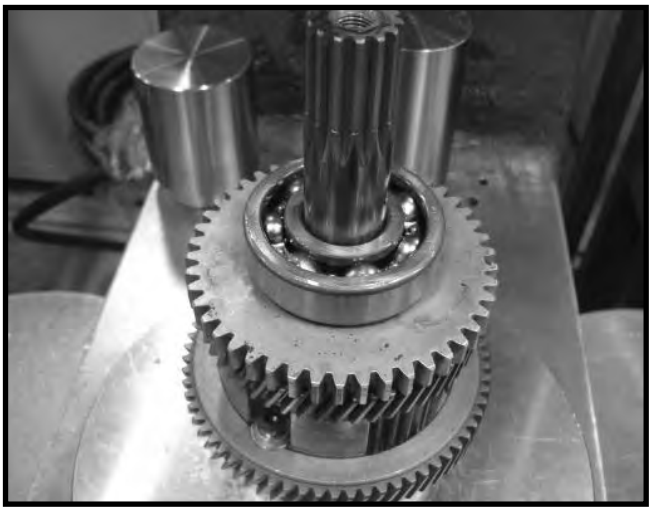
Install **112-64 Bolts** with 2760 Loctite applied to the threads of the fasteners as well as the internal threads of the aluminum cage to which they will be installed. Torque to 35 ft-lbs





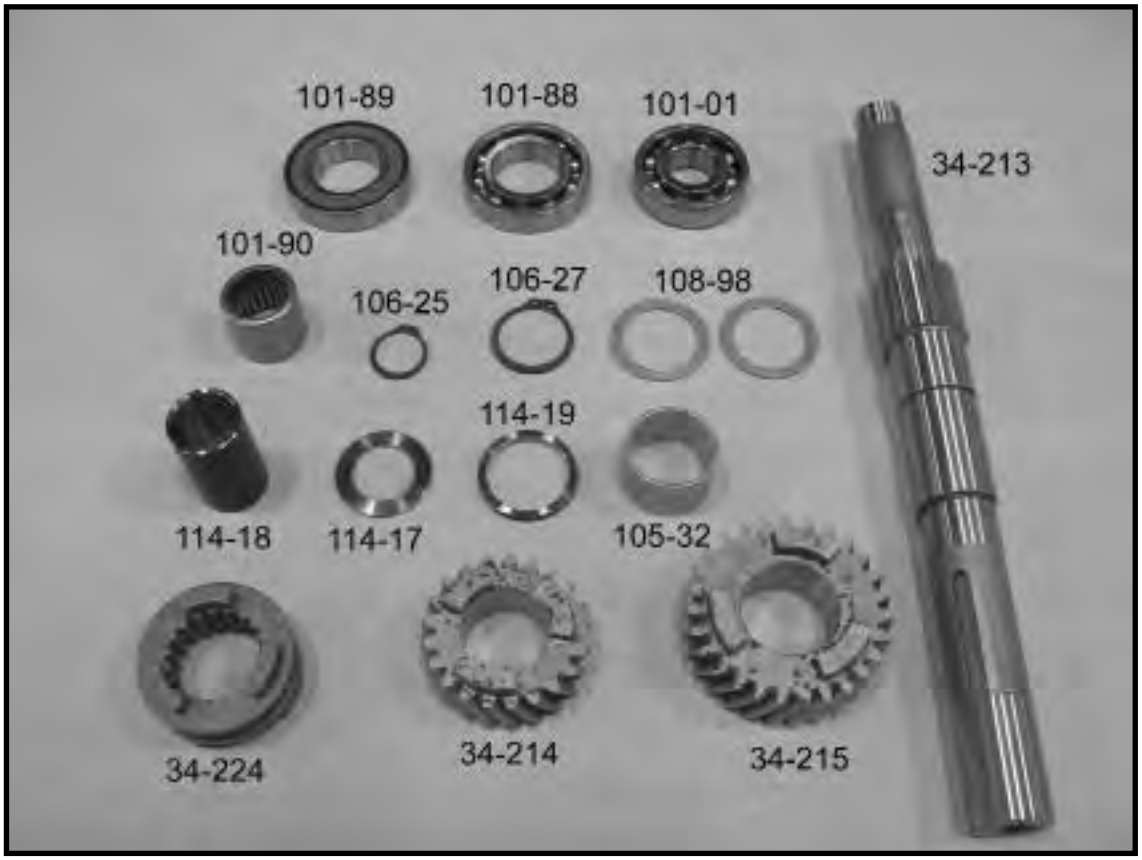
Install **108-94 Washer** onto the end of the **34-201 Brake Shaft** ***Photo 22*** followed by a **34-203 Spur Gear**, **114-16 Washer** and press on a **101-40 Ball Bearing.**





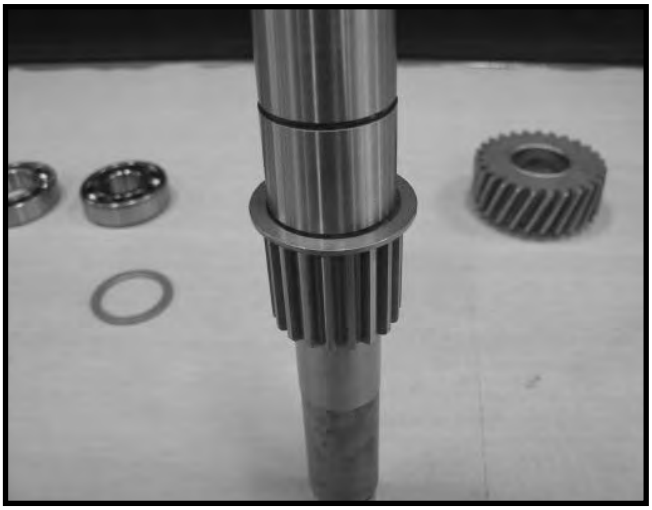
Repeat for the other side of differential assembly.

INPUT SHAFT ASSEMBLY



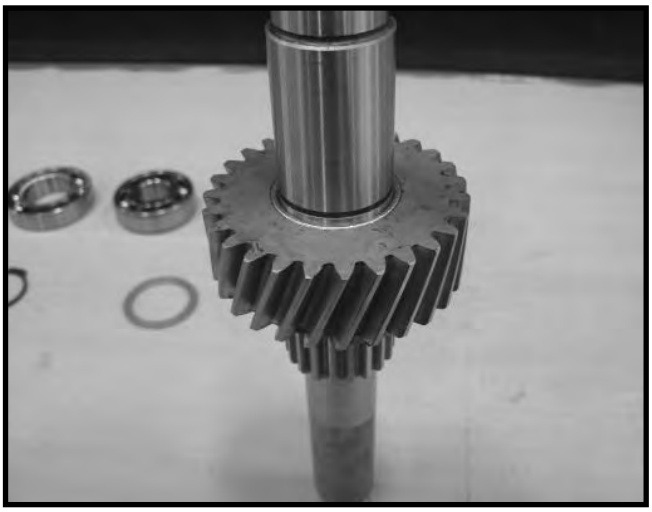
Components
101-01 Ball Bearing, 101-88 Ball Bearing, 101-89 Ball Bearing 101-90 Needle Bearing, 105-32 Bushing, 106-25 Retaining Ring, 106-27 Retaining Ring, 108-98 Washer, 114-17 Washer, 114-18 Spacer, 114-19 Spacer, 34-213 Input Shaft, 34-214 Reverse Pinion, 34-215 Forward Pinion, 34-224 Shift Collar

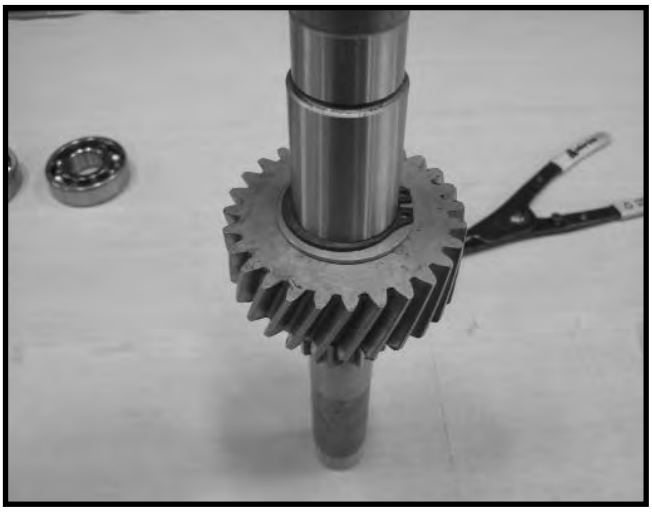




Press **105-32 Bushing** into **34-215 Forward Pinion**.

Install a **108-98 Washer** onto left side of input shaft.





Slide the **34-215 Forward Pinion** onto left side of input shaft.

Install another **108-98 Washer** onto left side of input shaft followed by a **106-27 Retaining Ring**.





Press **101-88 Ball Bearing** on next.

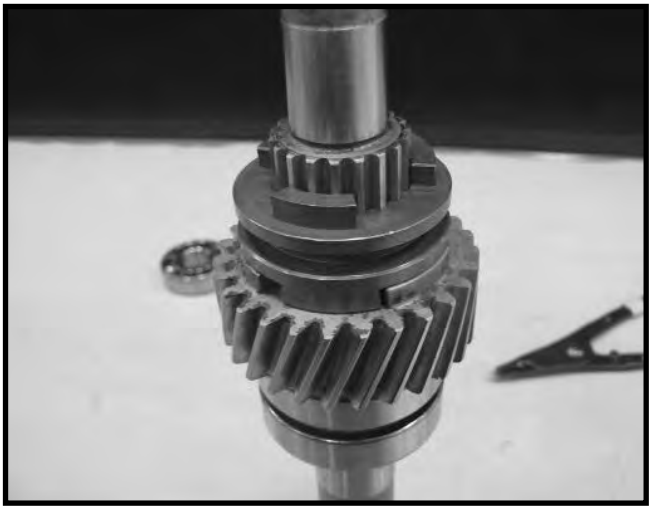
Install **114-19 Spacer**. Press **101-89 Bearing** on next.

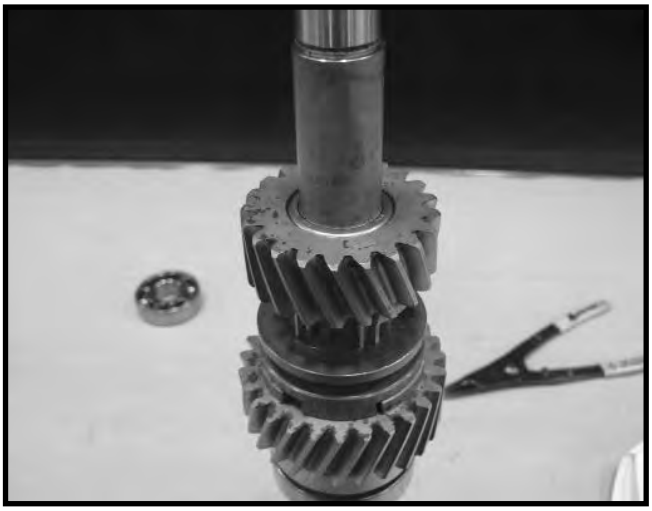
Note: **When pressing on 101-88 and 101-89 Bearings hold shaft assembly in press on the 34-215 Pinion.**





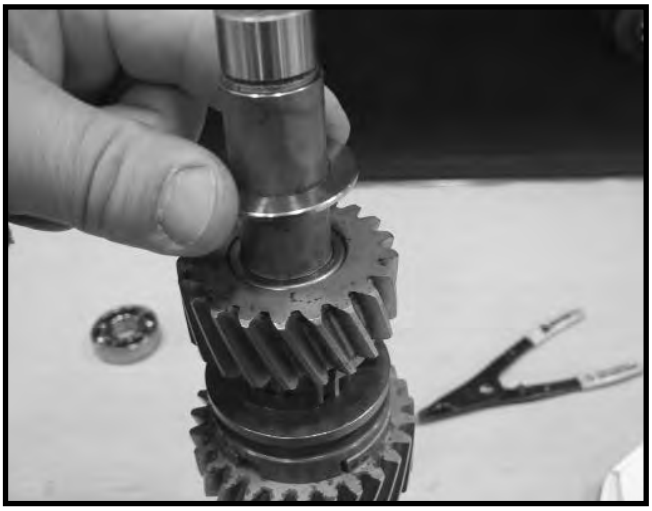
Press **101-90 Needle Bearing** into the **34-214 Reverse Pinion.**





Install **34-224 Shift Collar** on right side of input shaft. ***Note:*** **Shift collar must slide freely on shaft and fully engage with forward pinion.**

Install **34-214 Reverse Pinion,** followed by the **114-17 Washer**, **114-18 Spacer**, and a **106-25 Retaining Ring.**



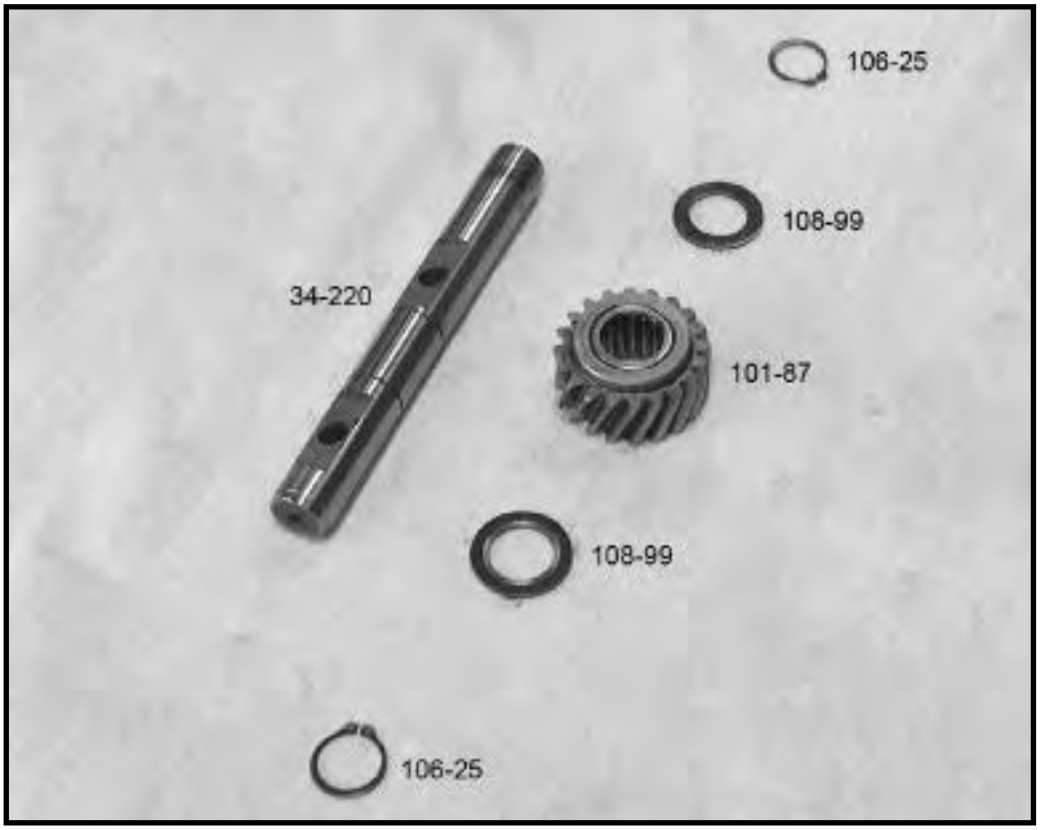






Press **101-01 Bearing** onto shaft.

REVERSE IDLER SHAFT ASSEMBLY



Components
101-87 Needle Bearing, 106-25 Retaining Ring, 108-99 Thrust Washer, 34-220 Idler Pinion, 34-221 Shift Shaft Reverse

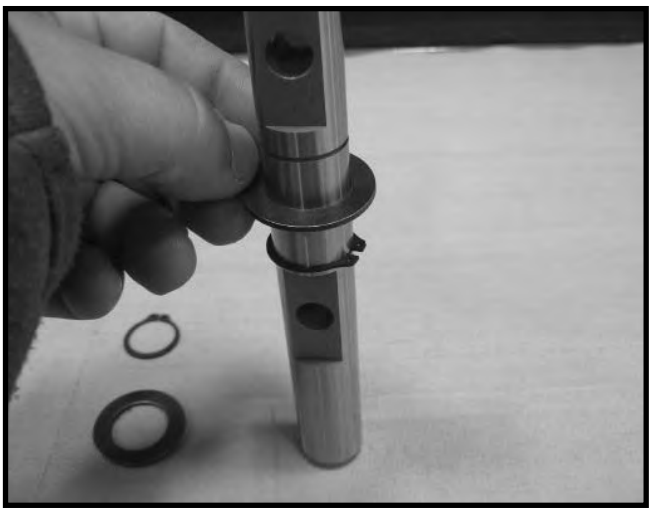




Press **101-87 Needle Bearing** into the **34-220 Idler Pinion.**

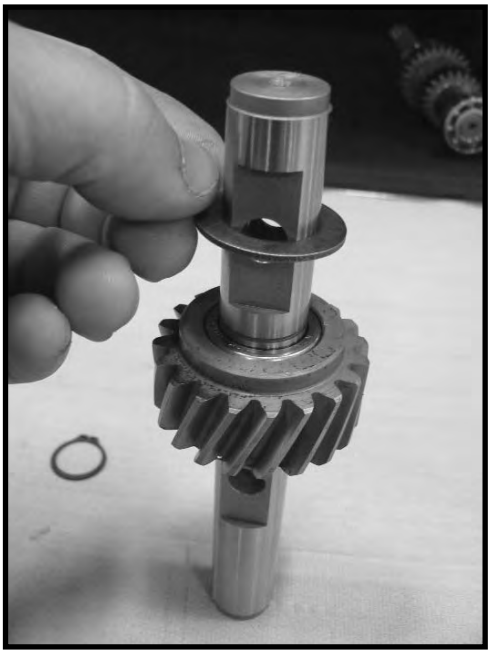
Install (1) **106-25 Retaining Ring** into a groove, then slip on a **108-99 Thrust Washer.**

Slide on the **34-220 Idler Pinion** that has the **101-87 Needle Bearing** installed on to shaft. ***Note:*** **Step on 34-220 Idler Pinion must face towards the end of the 34-221 Shaft that is the shortest.**





Slide on another **108-99 Thrust Washer,** and install a **106-25 Retaining Ring.**





IDLER SHAFT ASSEMBLY



Components: 108-96 Spacer, 34-209 Helical pinion, 106-28 Retaining ring, 34-210 Helical Pinion, 105-31 Bronze Bushing, 108-98 Thrust Washer, 34-223 Shift Collar, 34-208 Idler Shaft

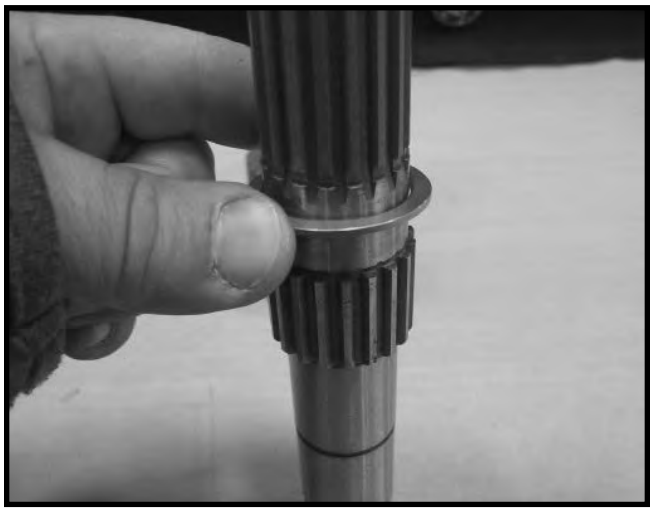
Press **105-31 Bronze Bushing** into **34-210 Helical Pinion.**

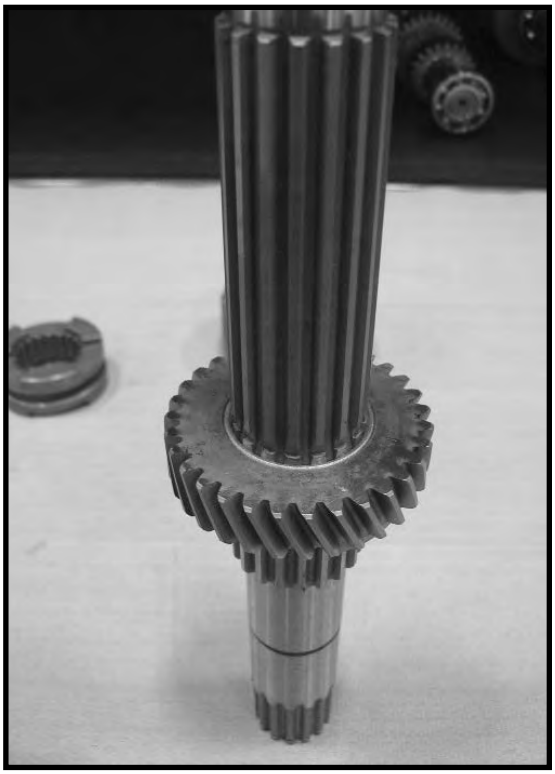




Install **108-98 Thrust Washer** onto left side (longer side) of **34-208 Idler Shaft**.

Slide **34-210 Helical Pinion** on the shaft.

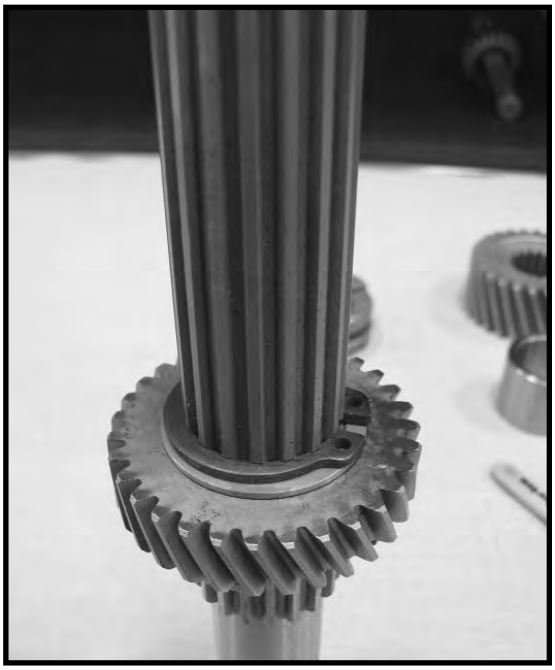




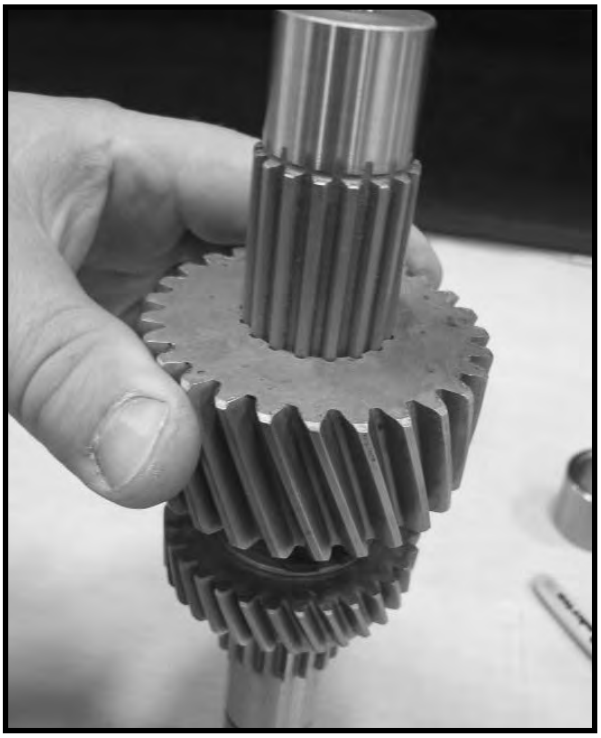
Install another **108-98 Washer.**

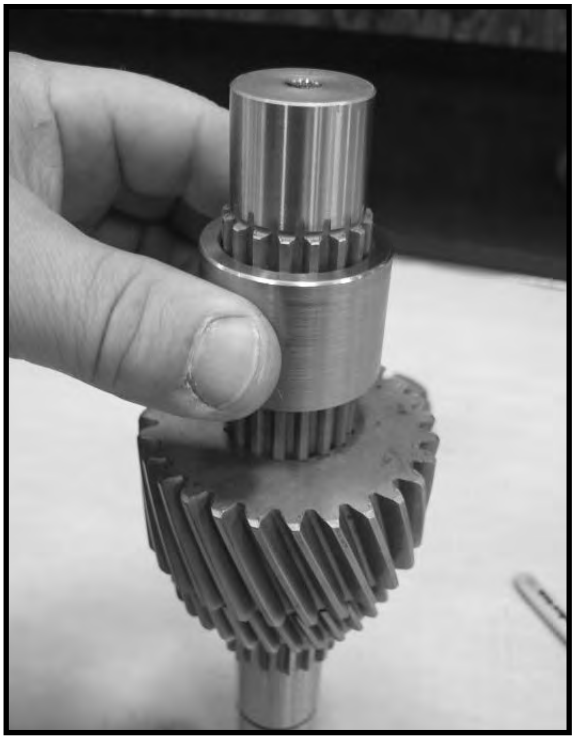
Install the **106-28 Retaining Ring.**



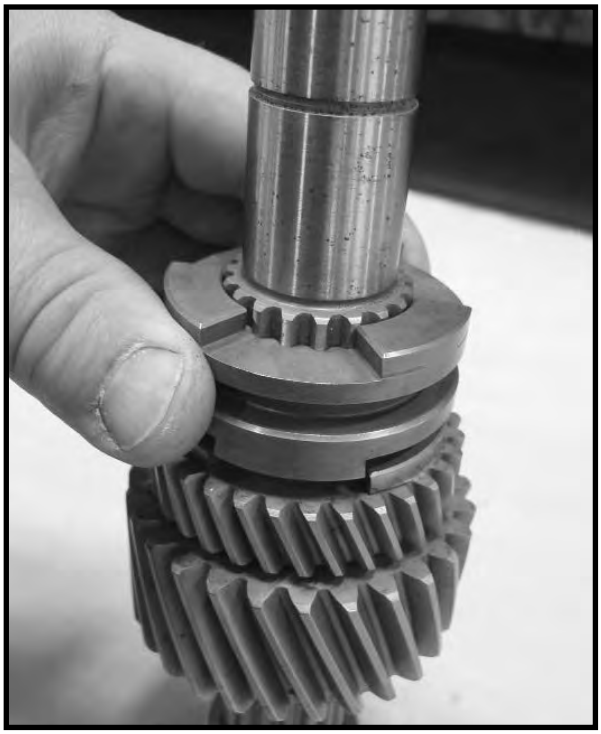


Slide **34-209 Helical Pinion** onto the idler shaft**,** followed by the **108-96 Spacer.**





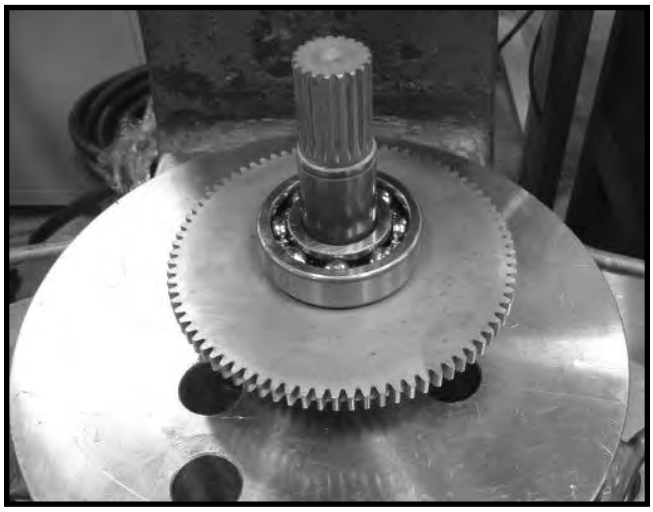
Slide the **34-223 Shift Collar** onto the idler shaft. ***Note:*** **Shift Collar must slide freely on idler shaft and not get tight.**



HOUSING COVER ASSEMBLY

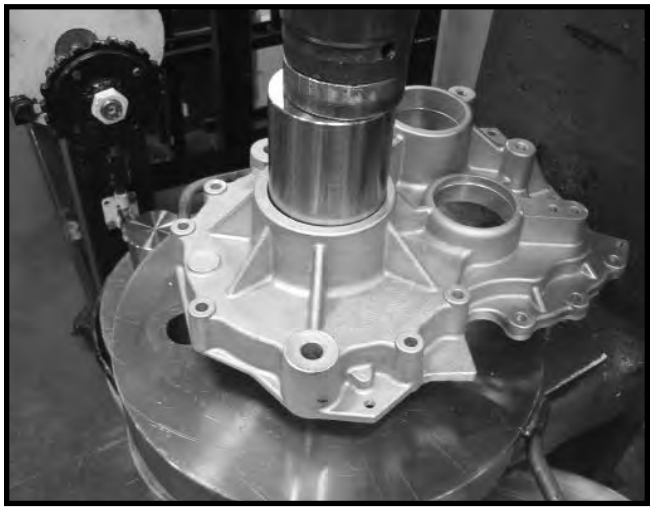
Put the **Right Output Shaft Assembly** on the press.

Install a **106-24 Retaining Ring** into the bottom groove in the **34-228 Cover** and **106-05** in the bottom groove of the upper differential bore in the cover.







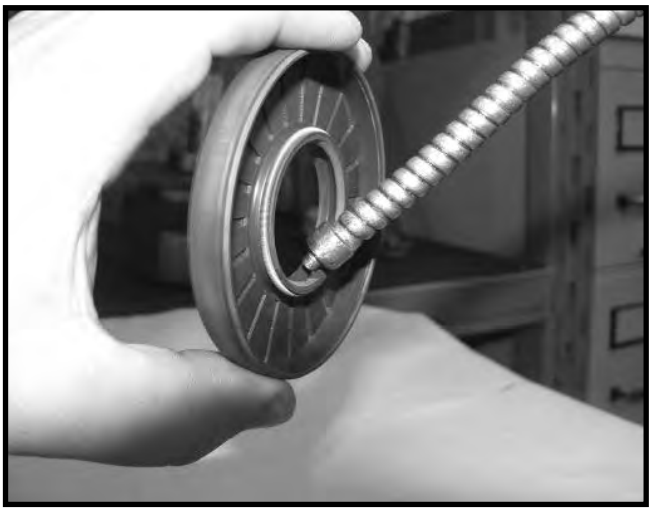


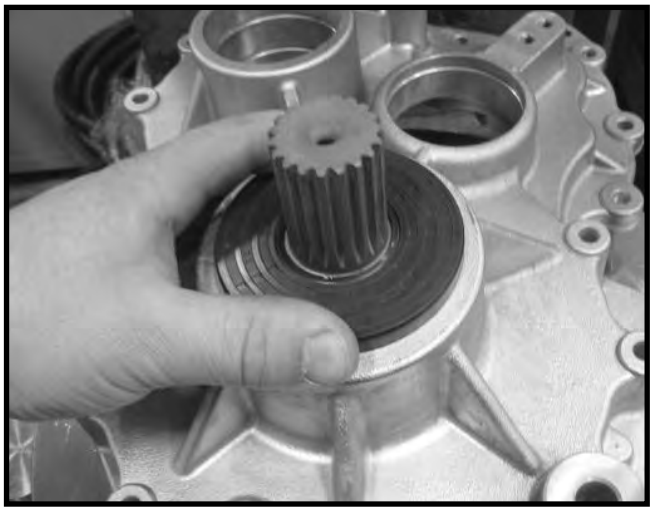
Press the **34-228 Cover** onto the Output Shaft assembly.

Put a small amount of oil on the inside of the **102-33 Oil Seal** and put seal on end of shaft and press down.

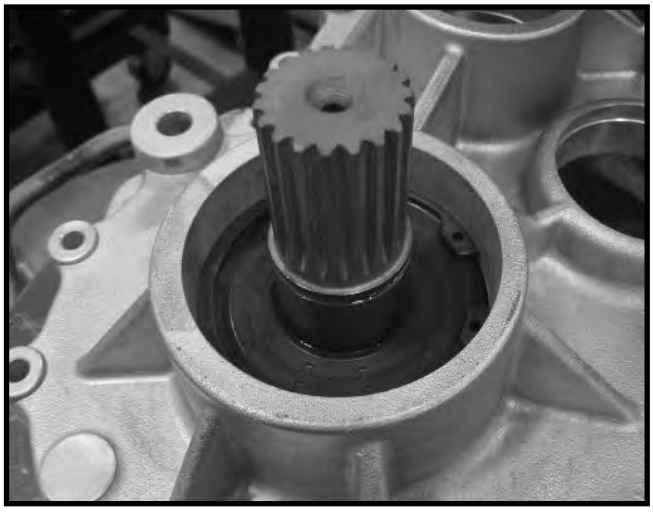
Note: **Make sure oil seal spring is facing the inside of the transmission. To insure that spring stays on seal, press seal on shaft as far as possible by hand.**

Install **106-24 Retaining Ring** into second groove of housing right in front of the oil seal**.**





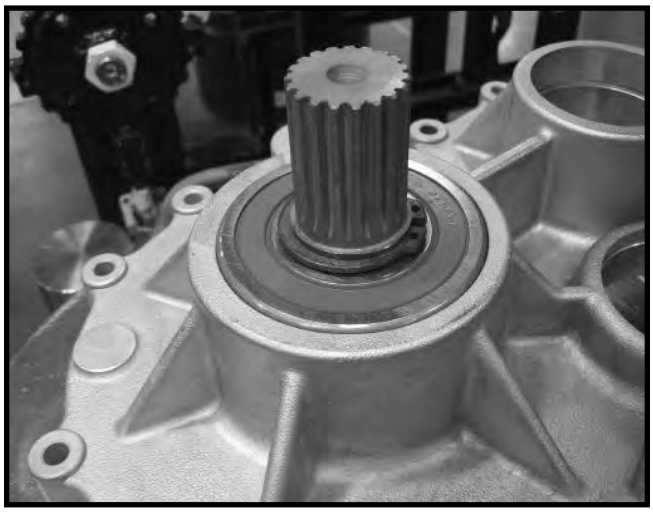




Press **101-81 Ball Bearing** onto housing and shaft.

Install **106-23 Retaining Ring** onto the output shaft.





HOUSING PRE-ASSEMBLY

Components
101-81 Ball Bearing, 102-33 Oil Seal, 106-05 Retaining Ring, 106-23 Retaining Ring, 106-24 Retaining Ring

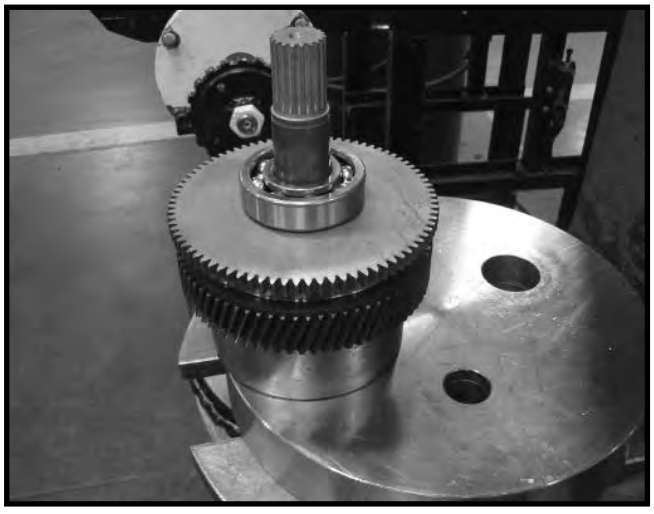
Install **106-24 Retaining Ring** into the largest hole opening on inside groove of housing. **.**

Install **106-05 Retaining Ring** into the small hole in the housing**,** followed by the smallest **106-26 Retaining Ring *3.***

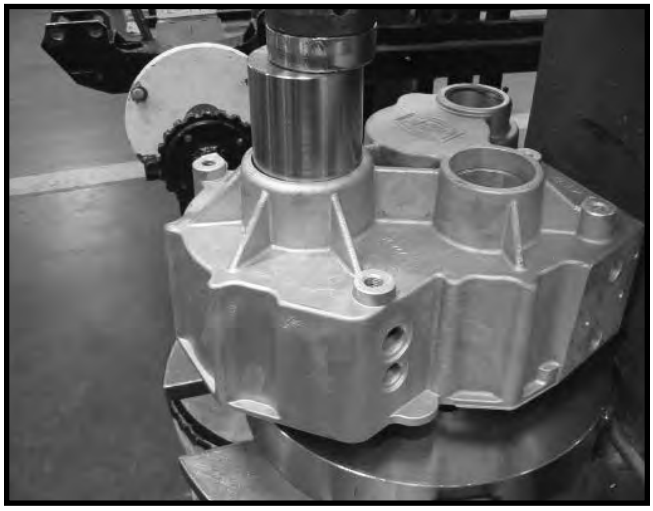


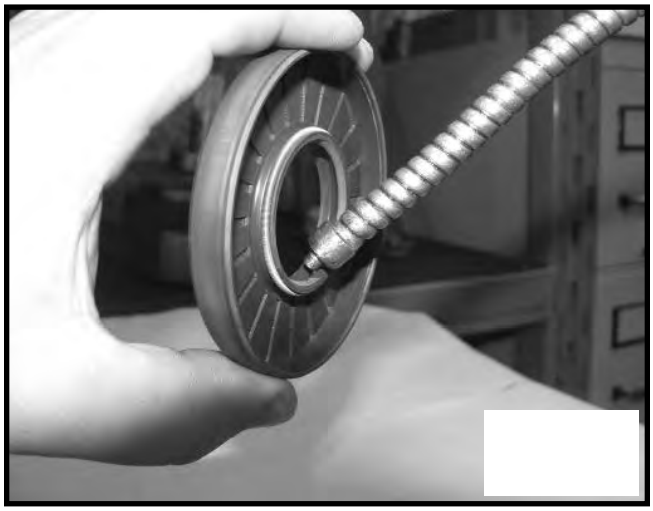






Set Planetary Hub assembly on press**,** and press the housing onto the Planetary assembly**.**





Put a small amount of oil on the inside of the **102-33 Oil Seal,** and put seal on end of shaft and press down.





Note: **Make sure oil seal spring is facing the inside of the transmission. To insure that spring stays on seal, press seal on shaft as far as possible by hand.**

Install **106-24 Retaining Ring** into second groove of housing right in front of the oil seal.

Press **101-81 Ball Bearing** onto housing and shaft.





Install **106-23 Retaining Ring** onto the output shaft.

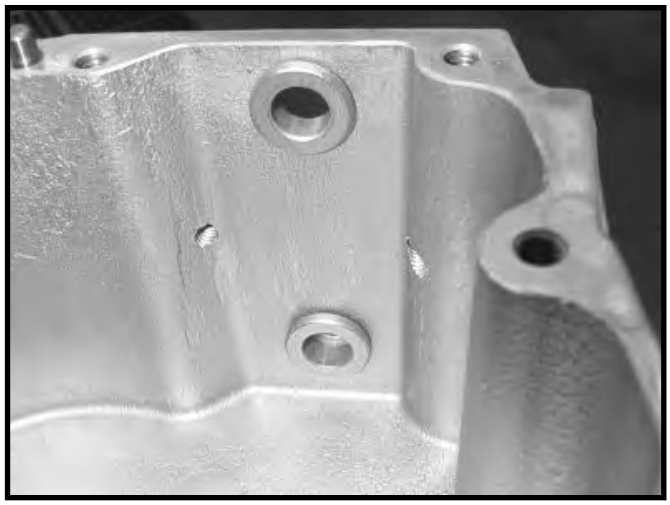


TRANSMISSION ASSEMBLY

Install **105-33 Flanged Bushing** into both holes in housing from the inside.

Install **104-03 Dowel Pins** into holes in the housing.

Install **101-73 Ball Bearing** into housing followed by **105-30 Flanged Bushing**.

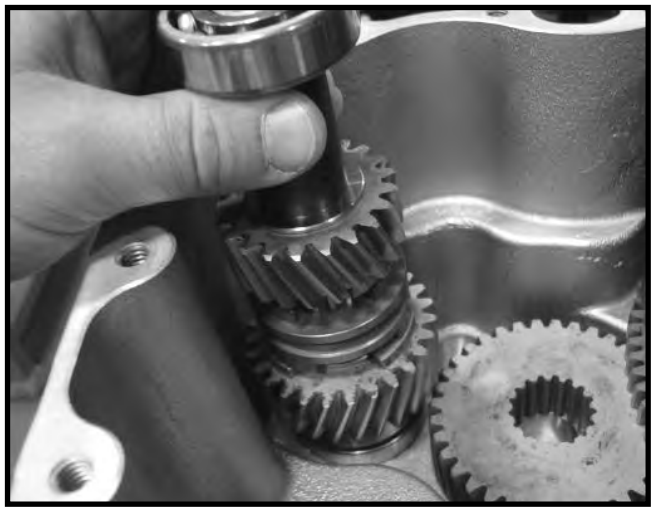


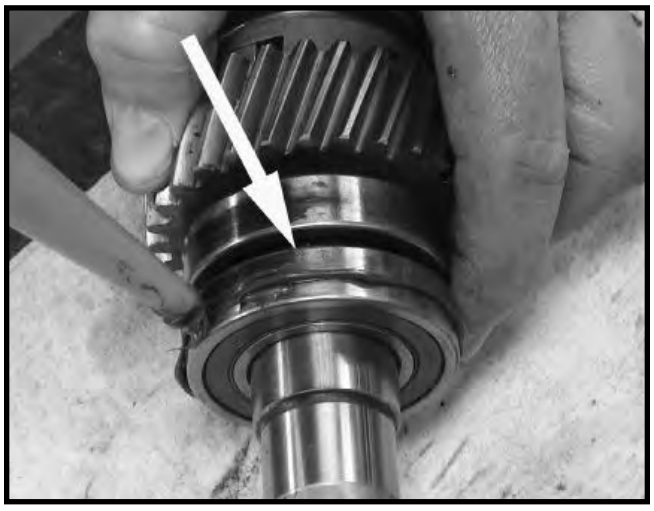


Place **34-211 Helical Gear** on top of the **105-30.**

Insert the input shaft assembly into the housing with the **101-01 Ball Bearing** facing upwards.

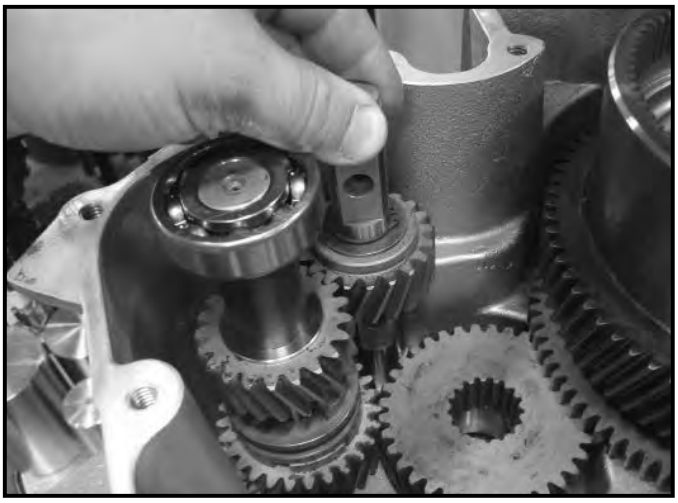
Put 598 Gasket Maker on first bearing input shaft before installing**.**

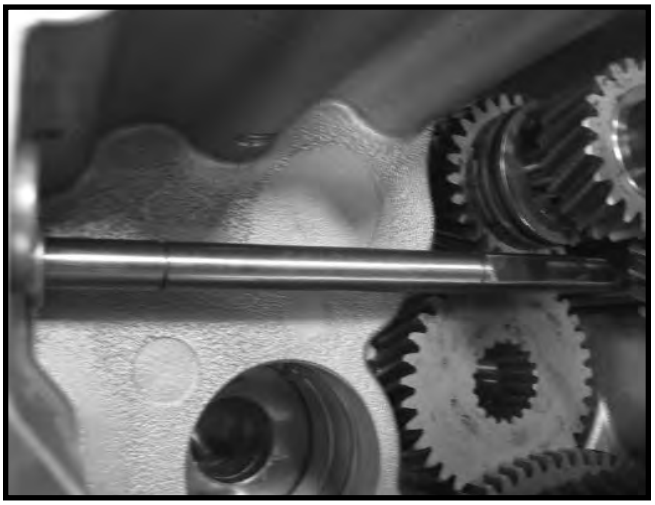




Note: **If input shaft needs to be pressed into housing you must use a special fixture that pushes on the 34-215 Forward Pinion or else transmission may not spin freely.** Install reverse idler shaft assembly with the flats on the 3/8" holes facing towards the inside of the transmission.

Note: **Step on 34-220 Idler Pinion must face upwards.**





Insert a **34-231 Shift Shaft** through the **105-33 Flanged Bushing** and through the **34-221 Shift Shaft,** and make sure it spins freely and does not bind.
Note: **If shaft is tight to turn the 34-221 Shift Shaft must be rotated.**

Pull the **34-221 Shift Shaft** back out half way and slide on a **127-186 Spring** followed by (2) **12731594 Washers**, (1) **114-21 Spacer** and a **34-232 Shift Fork.**



